SIMATS ENGINEERING

CSA3703 – SOFTWARE TESTING

ASSIGNMENT 4

CONSTRUCT, COMPARE AND JUSTIFY TEST PLANNING,
TEST DOCUMENTATION AND TEST ORGANISATION

POWERPAY ELECTRICITY BILL PAYMENT AND CONSUMER SERVICE PORTAL

Submission detail	Information
Student Name	Srikesh K
Register Number	192321040
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Faculty Name	Dr Gnanajeyaraman Rajaram

EXECUTIVE SUMMARY

PowerPay is a proposed electricity bill payment and consumer-service portal for three lakh consumers. Its functions include registration and login, meter-reading upload, slab-based bill generation, online payment, complaint and outage reporting, and SMS receipts. The previous pilot exposed four process weaknesses: a wrong bill at exactly 500 units, duplicate payment recording, a mismatch between the tested and released builds, and missing evidence of how much testing was completed. These are not isolated coding concerns; they indicate weaknesses in test design, configuration control, reporting, policy, organisation and defect analysis.

This report compares the three prescribed techniques in each of the nine tasks, makes a reasoned selection, and states where the selection could fail. The original plan assumes a ten-week window, four testers, one test server, six modules, about 180 functional conditions and a ₹1,50,000 tool budget. Estimates are shown transparently. The changed scenario expands the population to twelve lakh consumers, introduces smart-meter uploads every 15 minutes, reduces the window to four weeks and the test team to three testers, and requires masked test data plus auditable results.

Key decision	Recommendation	Reason
Debugging	Systematic fault localisation with regression tests	Repeatable evidence, minimal failing input and controlled version comparison.
Test plan	Complete plan with risk-based breakdown	Supports approval, audit and targeted change control.
Test item identity	Tagged baseline plus requirement traceability	Prevents tested-build/released-build mismatch.
Reporting	Daily dashboard + incident reports + level summary	Balances timely action, reproducibility and release readiness.
Policy/team	Formal measurable policy; module-aligned collaboration with independent release gate	Makes rules enforceable without removing domain knowledge.
Documentation	Risk-tailored hybrid	Full evidence for billing/payment; lean tool-generated records for lower-risk functions.
Changed scenario	Redesign around CI, masking, central evidence and staged gates	Original plan cannot fit the revised effort and audit constraints unchanged.

Academic integrity note: code, sample logs, defect times, pass/fail outcomes, effort estimates and reproducibility figures in this report are labelled as illustrative/simulated unless a real execution record is explicitly supplied. Before submission, replace simulated observations with actual exercise results where required by the instructor.

REPORT MAP Contents and task allocation

Section	Topic	Marks	Planned pages
Task 1	Testing vs debugging; fault localisation	4	6–12
Task 2	Test planning and plan components	7	13–22
Task 3	Test items, attachments and traceability	5	23–30
Task 4	Test reporting and incident evidence	6	31–38
Task 5	Testing policy development	7	39–46
Task 6	Test team organisation	4	47–52
Task 7	Skills of a test specialist	5	53–59
Task 8	Integrated test process and document sets	4	60–65
Task 9	Re-evaluation and redesign	8	66–74
Appendices	SRS, sample artefacts, checklists and evidence templates	—	75–80

The report follows the assignment's nine-task structure. Each task begins with a prediction, describes a controlled evaluation method, compares all three candidate techniques, provides task-specific tables or artefacts, and closes with a recommendation, limitations and what may happen if the selected control is omitted.

The evidence design is intentionally practical: a Chief Engineer should be able to see scope, status, risks and release recommendation quickly, while a developer should be able to reproduce a defect and an auditor should be able to trace a result to the exact build, test case and requirement.

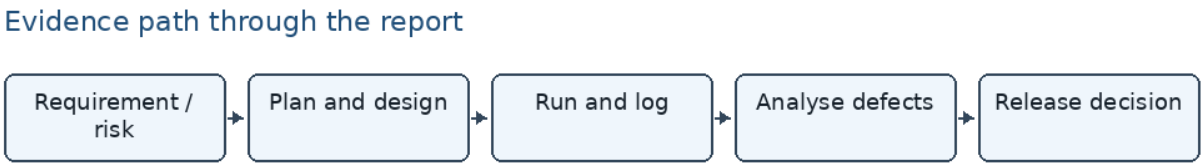


Figure: Evidence path through the report (illustrative process diagram).

BASELINE SRS PowerPay requirements – functional requirements

The following SRS excerpt is authored for this academic exercise because a faculty-provided SRS was not attached in the prompt. Requirement IDs are stable identifiers for planning and traceability; the Corporation must confirm tariff rules, authentication policy, payment-provider contracts, outage workflow and statutory data-retention rules before treating this excerpt as approved.

ID	Requirement statement	Risk	Acceptance idea
PP-R01	Consumers shall register with a verified consumer number and contact channel.	High	Valid/invalid registration cases; no duplicate account.
PP-R02	Authenticated consumers shall sign in and securely sign out.	High	Correct role access; session expires according to policy.
PP-R03	The portal shall accept meter readings with timestamp, meter ID and validation status.	High	Valid upload accepted; malformed or duplicate reading rejected.
PP-R04	The portal shall calculate energy charges using the approved slab tariff.	Critical	Boundary tests at 100, 300, 500 and values adjacent to each boundary.
PP-R05	The portal shall show bill units, slab breakdown, taxes, fees and total separately.	High	Displayed components sum to the final amount.
PP-R06	A bill shall have a unique consumer ID, billing period and bill reference.	High	No duplicate active bill for the same period.
PP-R07	A payment request shall include bill reference, amount and unique idempotency key.	Critical	Repeated request does not create a second payment.
PP-R08	Payment status shall be reconciled with the payment gateway before receipt issuance.	Critical	No receipt for unconfirmed payment.
PP-R09	Successful payment shall generate a unique receipt and transaction reference.	High	Receipt amount and reference match ledger.
PP-R10	The portal shall send an SMS receipt to the verified contact after success.	Medium	One notification per eligible successful payment.
PP-R11	Consumers shall raise and track a complaint with a unique complaint ID.	Medium	Status, timestamps and ownership are visible.
PP-R12	Consumers shall report an outage with location and incident details.	Medium	Duplicate reports can be linked; urgent reports are routed.
PP-R13	The portal shall support at least 3 lakh consumers in the initial release.	High	Load test profile agreed and measured.
PP-R14	Privileged staff actions shall be access-controlled and auditable.	High	Unauthorised action denied; audit event recorded.

BASELINE SRS Non-functional requirements and assumptions

ID	Requirement / assumption	Verification approach	Owner
PP-N01	Search, bill view and payment status response should meet a defined service target under agreed load.	Load test with percentile metrics; target must be approved before execution.	Chief Engineer / IT
PP-N02	Payment operations shall be idempotent and reconcilable after timeout or retry.	Duplicate, timeout, delayed callback and restart tests.	Payment developer / tester
PP-N03	Test datasets shall not expose real consumer personal data without authorised lawful basis.	Use synthetic or masked datasets; inspect logs and attachments.	Data owner / test manager
PP-N04	The exact tested artifact shall match the candidate release artifact.	Hash, tag, build ID and deployment manifest comparison.	Build engineer
PP-N05	Every executed test result shall be traceable to test ID, build, environment and timestamp.	Tool export and audit sampling.	Test manager
PP-N06	Critical defects shall block release unless formal risk acceptance is permitted by law and policy; for billing/payment criticals, no waiver is recommended.	Severity review and release gate.	Release authority
A-01	Tariff examples in this report are illustrative and must be approved by the billing-section coordinator.	Rule walkthrough and signed tariff table.	Billing coordinator
A-02	The original team consists of 13 people: 1 test manager, 4 testers, 5 developers, 1 build engineer, 1 business analyst and 1 billing coordinator.	Capacity plan.	Project manager
A-03	One shared test server is available; contention and reset time must be included in schedule risk.	Environment booking log.	Build engineer
A-04	Tool budget is ₹1,50,000; free/open-source tools may reduce licence cost but not configuration and training effort.	Costed tool shortlist.	Test manager

Assumptions requiring approval: tariff boundary semantics (whether each rate applies to all units or only units inside each slab), rounding rule, tax order, treatment of corrected readings, payment settlement semantics, complaint service targets, outage priority, consumer-data retention and performance service-level objectives. A test plan must not silently invent these business rules.



Figure: Requirement-to-test evidence (illustrative process diagram).

TASK 1.1 Testing and debugging goals

Prediction: a systematic workflow will leave stronger evidence than ad hoc print statements. Testing begins by executing a defined test and ends with a recorded pass/fail result; debugging begins after a failure is observed and ends when the cause is understood, fixed and regression-tested.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Print/log statements	Easy to add and inspect; output may be noisy and may change timing. Logs can miss hidden state or require repeated edits.	Useful for simple calculations and production-safe diagnostic events; not sufficient alone for the 500-unit boundary.
B. Interactive debugger	Breakpoints, stepping and watches expose control flow and variable values; depends on reproducible local setup and skill.	Effective for inspecting slab index, taxable units and rounding at a single failing input.

C. Systematic localisation	Minimise failing input, compare versions, bisect changes, record hypothesis and add regression test.	Best evidence trail and repeatability; requires version history and discipline.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Time to locate	Timer from reproducible failure to root cause; repeat each method on the same seeded fault.	Compare medians across four faults; do not generalise from one run.
Runs	Count complete test reruns plus diagnostic executions.	Fewer runs may be faster, but correctness and evidence matter too.
Evidence	Capture log excerpt, debugger variable snapshot or bisect range and regression ID.	Auditor can follow the chain from failure to fix.

Illustrative test artefact

```
# Illustrative cumulative-slab calculation; confirm tariff semantics first
def energy_charge(units, slabs):
    if units < 0:
        raise ValueError("units cannot be negative")
    remaining, total = units, 0
    for width, rate in slabs:
        used = min(remaining, width)
        total += used * rate
        remaining -= used
        if remaining <= 0:
            break
    return total
```

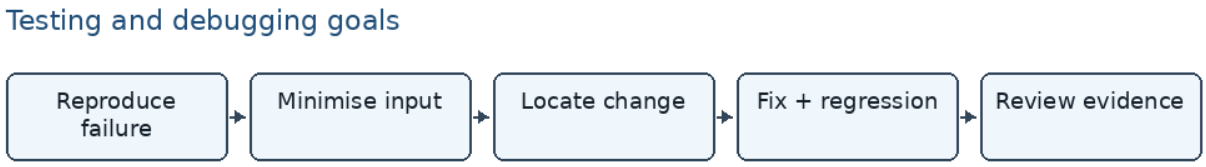


Figure: Testing and debugging goals (illustrative process diagram).

Recommendation and limitation

Require systematic localisation for billing/payment defects; allow logs and debugger as supporting tools. The systematic method may be slower for a trivial typo, but it is repeatable and protects against an undocumented local fix. If omitted, a patch may repair 500 units while breaking 501 units or another slab boundary.

TASK 1.2 Seeded billing faults

Prediction: four seeded faults will show that no single tool automatically identifies every root cause. Faults must be introduced in a controlled branch and their IDs hidden from the tester until the run is complete.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Fault F1: off-by-one	Condition uses units < 500 where the approved boundary requires <= 500 (or vice versa).	Test 499, 500 and 501 units; compare slab breakdown.
Fault F2: wrong rate	Third slab rate is ₹7.50 instead of ₹7.00.	Use a value fully inside the third slab and calculate expected charge independently.

Fault F3: double counting	Units in a previous slab are charged again in next slab.	Compare component sum with independently calculated expected amount.
Fault F4: rounding order	Round each slab before summing when policy requires final-total rounding.	Use decimal values near a half-paisa/paise boundary if applicable.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Reproducible input	Fixed tariff version, meter units and consumer class.	All approaches begin from same test data.
Fault masking	Run tests around each boundary and across multiple slab transitions.	One fault may hide another; test faults separately first.
Root-cause proof	Link failing assertion to code line/commit and fix.	A plausible explanation is not sufficient without regression evidence.

Illustrative test artefact

```
# Example regression boundary set
for units, expected in [(499, expected_499), (500, expected_500), (501, expected_501)]:
    assert energy_charge(units, approved_slabs) == expected
```

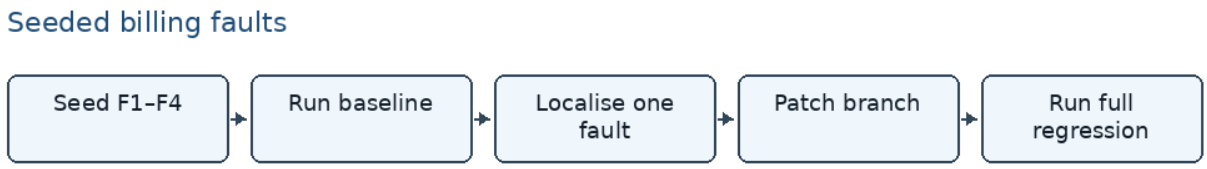


Figure: Seeded billing faults (illustrative process diagram).

Recommendation and limitation

Record method-specific time and run counts during the exercise. If no actual timer is used, enter 'not measured' rather than inventing exact seconds. The result table should identify method, fault ID, elapsed minutes, runs, tool, root cause, regression ID and reviewer.

TASK 1.3 Regression evidence before and after fix

Prediction: each fix should be paired with a test that fails on the faulty version and passes on the corrected version. A test that passes both versions does not demonstrate that it protects against the seeded defect.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Before fix	Run the focused test against the seeded-fault commit and preserve failure output.	Expected: FAIL for the targeted fault.
After fix	Run the same test on the fix commit; run nearby boundaries and existing regression set.	Expected: PASS without unrelated failures.
Review	A second person reviews expected value and code change independently.	Prevents a test from simply copying the faulty implementation.
Reopen prevention	Check neighbouring slabs, rounding, tax and bill total; rerun after merge.	Reduces chance of fix-induced regression.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Regression strength	Fault-specific test fails before and passes after.	Strong evidence that the test detects the fault.
Collateral damage	All related boundary and calculation cases pass after patch.	Protects adjacent requirements.
Review quality	Independent expected-value calculation signed by billing coordinator.	Avoids shared misunderstanding of tariff semantics.

Illustrative test artefact

```
# Pseudocode for before/after evidence
result_before = run_case(build="faulty-build", case="BILL-500")
result_after  = run_case(build="fixed-build", case="BILL-500")
assert result_before.status == "FAIL"
assert result_after.status == "PASS"
```



Figure: Regression evidence before and after fix (illustrative process diagram).

Recommendation and limitation

Every fix must include a regression test and review. If a fix introduces a new fault, classify it separately, preserve both the original and new test, and determine whether the issue arose from incomplete impact analysis, weak expected values, or an unsafe change. The pilot's 18% reopen rate suggests fixes were not consistently validated or reproduced.

TASK 1.4 AI-assisted debugging critique

Prediction: an AI assistant can suggest hypotheses quickly but may assume cumulative or flat slab charging incorrectly. Its output is a proposal, not evidence, and must be checked against the approved tariff specification.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Explanation check	Ask AI to explain 499/500/501 behaviour and identify assumptions.	Reject any explanation that does not state slab interpretation.
Patch review	Inspect boundary operators, loop termination, decimal arithmetic and invalid inputs.	Look for fixes that shift the defect to 501 units or alter unrelated slabs.
Regression validation	Execute all four seeded-fault tests plus existing bill and payment tests.	Only observed test results support acceptance.
Privacy/security	Use synthetic tariff and consumer data; do not paste real account identifiers into public tools.	Protect consumer data and comply with policy.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Completeness	Does the patch address root cause and adjacent boundaries?	Human review plus focused regression.
Correctness	Does output match independently approved expected values?	Reference calculation, not model explanation.

Auditability	Is prompt, patch, reviewer and test run linked?	Retain tool use only if policy permits and evidence is adequate.

Illustrative test artefact

```
# AI-generated patch acceptance gate
assert requirement_id is not None
assert reviewer_approved_patch is True
assert boundary_suite.status == "PASS"
assert payment_regression.status == "PASS"
```



Figure: AI-assisted debugging critique (illustrative process diagram).

Recommendation and limitation

Use AI for hypothesis generation, test-data ideas and explanation review, but never for unreviewed production changes. The AI may fail when context is incomplete, tariff policy is ambiguous or its code silently changes rounding. If the output cannot be traced to a requirement and verified by tests, do not merge it.

TASK 1.5 Testing versus debugging workflow

Testing's purpose is to reveal failures against expected behaviour; debugging's purpose is to identify and remove the cause. They are connected but have different completion criteria and owners.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Testing start/end	Starts with approved test conditions and build; ends with result, evidence and defect report.	A failed test creates a debugging input; it does not by itself locate the fault.
Debugging start/end	Starts with a reproducible failure and evidence; ends with verified cause, fix and regression.	Closing a code change without rerunning tests is incomplete debugging.
Ownership	Tester owns objective failure evidence; developer investigates and fixes; tester/reviewer verifies.	Avoids treating 'works on my machine' as closure.
Defect lifecycle	New → triaged → assigned → fixed → retest → closed or reopened.	Every transition should have evidence and timestamps.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Handoff quality	Incident includes build, input, expected/actual result and logs.	Higher reproducibility reduces diagnosis delay.
Closure quality	Fix commit linked to a passing regression test.	Prevents unverified closure.
Reopen analysis	Group reopened defects by root cause.	Use data to improve review and training.

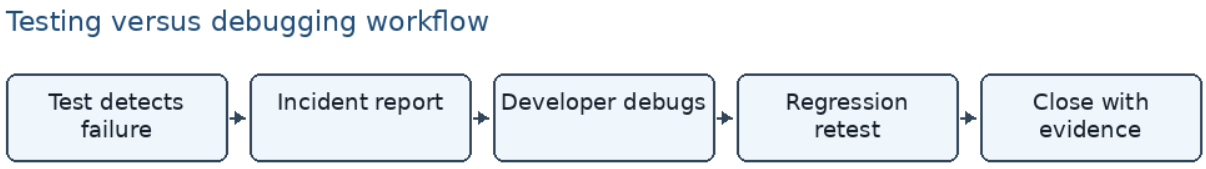


Figure: Testing versus debugging workflow (illustrative process diagram).

Recommendation and limitation

Testing is not responsible for proving the exact faulty line; debugging is not responsible for deciding overall release readiness. The test manager consolidates both streams. If roles are blurred, teams may close a defect after a code edit without checking the original failure or neighbouring behaviour.

TASK 1.6 Task 1 results and decision table

The final decision should be based on measured outcomes from the same four faults. The table below is a recording template; the illustrative ranking is a prediction, not a claimed experiment result.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Print/log	Low setup effort; diagnostic runs may be repeated.	Useful when intermediate values are obvious; can produce noisy output.
Interactive debugger	Moderate setup; precise state inspection.	Useful for branch and state errors; depends on reproducible local environment.
Systematic localisation	Higher discipline; strong repeatability and traceability.	Preferred for financial logic and defects likely to recur.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Measure	Record	Acceptance interpretation
Median time	Minutes across four faults per method.	Lower is better only if the root cause is correctly established.
Runs	Number of test/diagnostic runs.	Compare under same test environment.
Evidence completeness	0–5 checklist score for input, build, hypothesis, fix, regression.	Systematic method should leave the strongest audit trail.

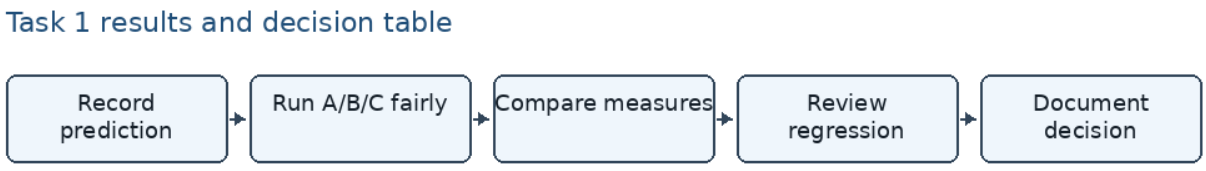


Figure: Task 1 results and decision table (illustrative process diagram).

Recommendation and limitation

Decision: systematic localisation is the required default for tariff and payment defects; logs and interactive debugging are complementary tools. A method can be faster on one fault but still be a weaker organisational standard if it leaves no reproducible evidence. Replace the predicted ranking with observed results from the lab exercise.

TASK 2.1 Plan structures and approval needs

A one-meeting approval requires an executive summary but the audit needs full detail. Prediction: a complete plan with a risk-based breakdown will best balance both needs.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Iteration-level agile	Short and easy to update; may omit suspension rules, item identification or approvals if used alone.	Useful for two-week delivery cadence but not sufficient as the only audit plan.
B. Complete IEEE 829-style	Covers scope, items, approach, criteria, deliverables, tasks, environment, roles, schedule, risks and approvals.	Strong approval baseline; can become cumbersome if copied without tailoring.
C. Risk-based test-level plan	Maps risk to level, technique, effort and exit condition.	Excellent prioritisation; must still include the mandatory administrative components.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Approval usability	Executive summary plus detailed annexes.	Chief Engineer sees decisions first; auditor can inspect details.
Completeness	Score each component present/partial/missing.	Missing build identity or exit criteria repeats pilot weaknesses.
Change handling	Record re-approval trigger for tariff revision.	Plan remains controlled as assumptions change.

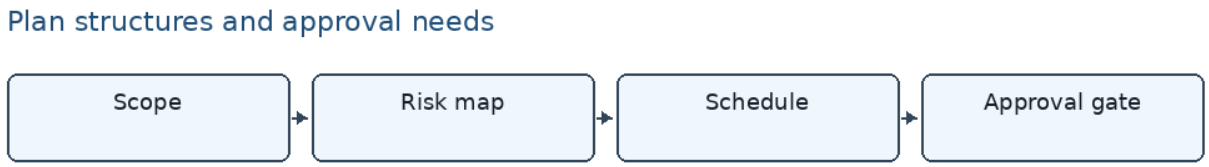


Figure: Plan structures and approval needs (illustrative process diagram).

Recommendation and limitation

Choose B as the controlled master plan, organised using C's risk/test-level mapping and supported by concise iteration sheets. This hybrid prevents omission while keeping the meeting-level view readable.

TASK 2.2 Plan A: iteration-level agile

Plan A is useful for a changing portal but is not a complete substitute for the master plan. Prediction: it will score well for update speed and lower for audit completeness.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Included	Iteration goal, selected stories, test data, risks, acceptance examples and exit condition.	Fast planning and visible work.
Often omitted unless mandated	Configuration baseline, suspension/resumption, test deliverables, formal approval, staffing/training and item pass/fail criteria.	Can recreate the tested-build mismatch and uncertain test progress.
Change management	Reprioritise work each iteration and link to backlog.	A regulator tariff change needs impact analysis and approval beyond the iteration card.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Setup effort	Estimate 2–4 hours per iteration sheet as a planning assumption.	Low document overhead.
Audit completeness	Checklist component score.	Partial scores require links to master policy/plan.
Failure mode	A sprint ends with tests passed on a non-release build.	Requires mandatory build identity and release manifest.

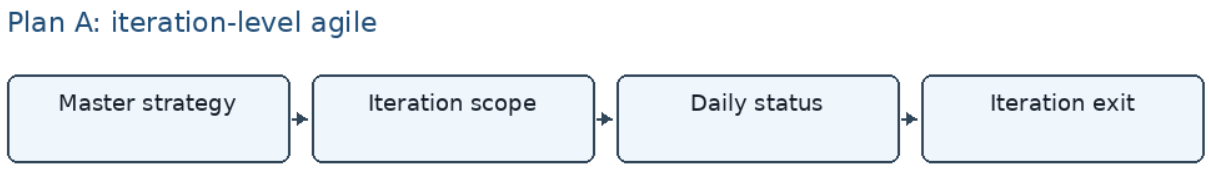


Figure: Plan A: iteration-level agile (illustrative process diagram).

Recommendation and limitation

Retain iteration sheets as child plans, not as the only approved document. The approach fails when each team interprets minimum content differently. Without a master plan, auditors cannot determine whether an omitted activity was intentionally excluded or simply forgotten.

TASK 2.3 Plan B: complete IEEE-style plan

Prediction: the complete plan will provide the strongest audit and approval basis, but only if the content is tailored and maintained.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Mandatory components	Test items; features in/out; approach; pass/fail; suspension/resumption; deliverables; tasks; environment; responsibilities; staffing/training; schedule; risks/contingencies; approvals.	Explicit accountability and criteria.
Potential weakness	Long generic sections may be ignored or stale.	Use concise tables, document owners and change history.
Approval design	Chief Engineer approves scope, critical criteria and residual risk; test manager owns detail.	Approval authority is explicit.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Completeness	Component checklist with present/partial/missing.	No missing critical control.
Maintenance	Review plan at each baseline and tariff change.	Change history prevents stale assumptions.
Usability	Executive summary and risk register at front.	Detailed content remains available for audit.

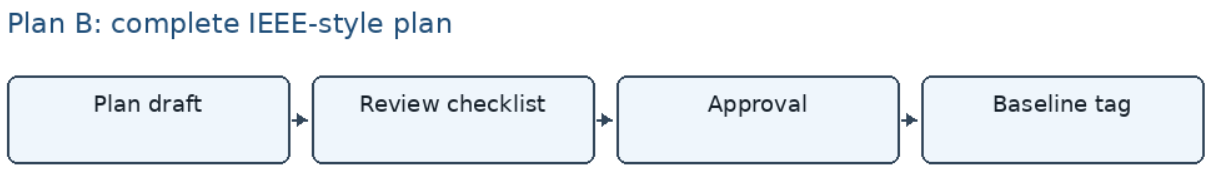


Figure: Plan B: complete IEEE-style plan (illustrative process diagram).

Recommendation and limitation

Use the complete structure, but keep it concise and link tool-managed details rather than duplicating all test cases in the plan. A long plan fails if teams cannot locate current versions or if changes are made without revision control.

TASK 2.4 Plan C: risk-based test-level breakdown

Risk-based planning links business impact to techniques and exit criteria. Prediction: this plan will prioritise financial integrity effectively but needs common administrative sections.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Critical risks	Incorrect tariff boundary, duplicate payment, receipt without confirmed payment, wrong release build and personal-data exposure.	Require focused automated and independent verification.
High risks	Bad meter data, bill total mismatch, complaint/outage loss, unavailable environment.	Require negative, interface, recovery and availability tests.
Lower risks	Noncritical layout variations and help text.	Sample by supported browsers and usability checks; do not omit accessibility or legal requirements.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Risk score	Probability × impact on 1–5 scale.	A 25 score is critical and receives highest priority.
Traceability	Each risk maps to requirement, test level, technique, owner and exit criterion.	Shows why effort is spent.
Residual risk	Record uncovered scenarios and decision authority.	No hidden exclusions.

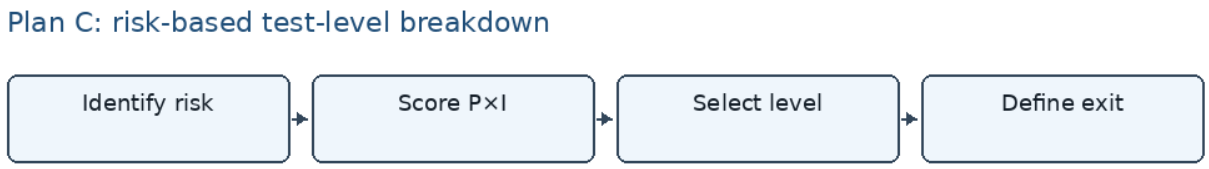


Figure: Plan C: risk-based test-level breakdown (illustrative process diagram).

Recommendation and limitation

Use C to structure test priorities inside B. Risk-based plans fail if probability and impact scores are arbitrary or if a low-likelihood catastrophic financial defect is down-ranked. The billing coordinator must validate business impact.

TASK 2.5 Effort calculation and feasibility

Prediction: the original four-testers/ten-week capacity is adequate for planned functional test work, but capacity must include leave, environment waiting, meetings and change requests.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Design effort	180 conditions × 0.4 h = 72 h.	Baseline design estimate.
Execution effort	180 × 0.25 h × 3 cycles = 135 h.	Three cycles include initial run and two repeat cycles as an assumption.

Retest effort	20% × (72 + 135) = 41.4 h.	Retest base explicitly defined.
Management overhead	15% × (72 + 135 + 41.4) = 37.26 h.	Applied after retest in this model.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Total planned effort	72 + 135 + 41.4 + 37.26 = 285.66 person-hours.	Approximately 286 hours.
Capacity	4 testers × 10 weeks × 30 h/week = 1,200 person-hours.	Gross capacity; not all available for test execution.
Capacity margin	1,200 − 285.66 = 914.34 hours before other activities.	Use margin for integration, performance, incidents and uncertainty.

Illustrative test artefact

```
design = 180 * 0.4
execution = 180 * 0.25 * 3
retest = 0.20 * (design + execution)
management = 0.15 * (design + execution + retest)
total = design + execution + retest + management
assert round(total, 2) == 285.66
```

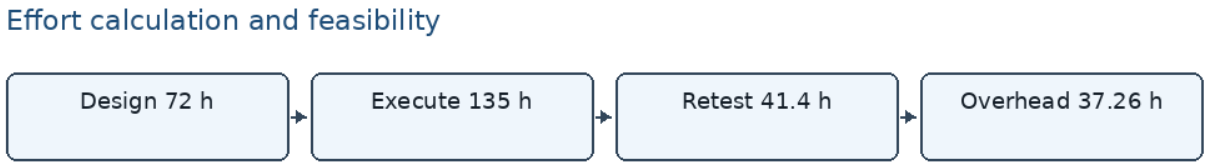


Figure: Effort calculation and feasibility (illustrative process diagram).

Recommendation and limitation

Feasibility: the arithmetic indicates the 180-condition functional effort fits within nominal capacity. It does not prove the entire ten-week project is feasible because system/load testing, setup, meetings, test data, defect triage, rework and leave must also be planned. Maintain a contingency reserve and update estimates weekly.

TASK 2.6 IEEE component checklist

The checklist maps plan content to the pilot-year failure modes. Scores should be supported by a link or section number, not a tick without evidence.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Present	The plan explicitly names the item, build, feature, approach, criterion, deliverable, environment, owner and approval.	Auditable and repeatable.
Partial	Information exists elsewhere but is not linked or is too vague to execute.	Risk of interpretation differences.
Missing	No approved statement or evidence exists.	Treat as a planning defect and close before relevant test work starts.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Item identity	Build ID, tariff table hash and deployment manifest.	Prevents tested/released mismatch.
Exit criteria	Critical defect threshold, pass rate by risk, traceability and evidence completeness.	Prevents unsupported readiness claims.
Suspension/resumption	Trigger and restart checklist for unavailable server or unstable build.	Prevents wasting effort on unreliable execution.

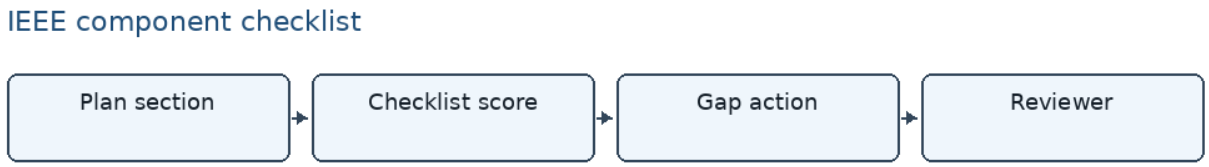


Figure: IEEE component checklist (illustrative process diagram).

Recommendation and limitation

Expected scoring: a short iteration plan alone is partial for most formal components; a complete IEEE-style plan should be present for all mandatory sections; a risk-based plan is strong for approach/risks/effort but may be partial for staffing, approvals and deliverables unless explicitly added. The final plan must contain all critical components.

TASK 2.7 Risk register and contingencies

Prediction: risks with high impact and high probability should have named owners and contingency actions before the project begins.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
R1 Environment unavailable	P=4, I=5, score 20. Owner: build engineer.	Reserve time slots; smoke-check daily; maintain deployable backup image and synthetic data.
R2 Late/unstable build	P=4, I=5, score 20. Owner: build engineer/development lead.	Enforce build entry gate; freeze release candidate; keep last known-good build.
R3 Tariff rule changes	P=3, I=5, score 15. Owner: billing coordinator.	Version tariff tables; impact analysis; re-approve expected values and regression set.
R4 Tester absence	P=3, I=3, score 9. Owner: test manager.	Cross-train; keep test procedures current; rebalance module ownership.
R5 Payment gateway instability	P=3, I=5, score 15.	Sandbox contract tests, virtual responses, retry/reconciliation scenarios.
R6 Masking/test-data defect	P=2, I=5, score 10.	Synthetic-first data; privacy review; access restrictions and log scans.
R7 Scope growth	P=4, I=4, score 16.	Change-control board and explicit effort re-estimation.
R8 Defect reopen	P=3, I=4, score 12.	Regression attachment, reviewer check and reopen root-cause analysis.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Highest contingency	R1, R2 and R7/R3 based on score and impact.	Escalate immediately and protect critical billing/payment tests.

Review frequency	Daily for blockers; weekly for full risk register.	Risk score changes as evidence arrives.
Trigger	Server unavailable > half-day, candidate build misses cutoff, tariff change approved.	Apply named contingency and record impact.

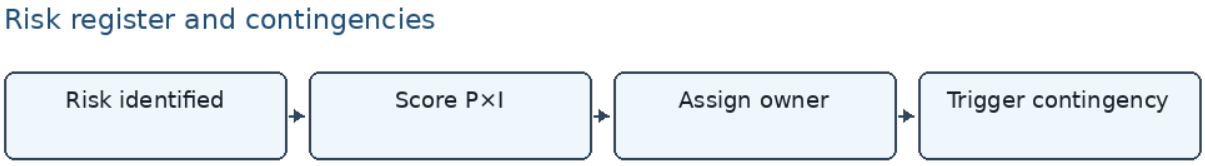


Figure: Risk register and contingencies (illustrative process diagram).

Recommendation and limitation

Probability and impact are planning estimates, not observed historical probabilities. The team should validate them at kickoff and revise weekly. A risk register without a contingency, trigger and owner is only a list of worries.

TASK 2.8 Tariff change in week 6

Prediction: a regulator tariff revision is a baseline change, not just an extra test case. It affects requirements, expected results, test data, bill examples, traceability and approval.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Iteration plan response	Add a story and update the current iteration scope.	Fast, but may miss dependent documents.
IEEE plan response	Raise change request; update version, scope, schedule, risk, criteria and approvals.	Full governance and audit trail.
Risk-based response	Re-score billing risks and prioritise boundary and historical regression cases.	Protects the highest-risk area quickly.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Re-approval needed	Approved tariff rule, test design, expected values, impacted cases, release criteria and schedule.	Billing coordinator and test manager sign; Chief Engineer approves material scope/time change.
Regression set	All tariff boundaries, sample bills, payment amount consistency, receipt and refunds if impacted.	Prevents stale expected amounts.
Evidence	Old/new tariff version, effective date, test build, diff and rerun report.	Explains which tariff each result validates.

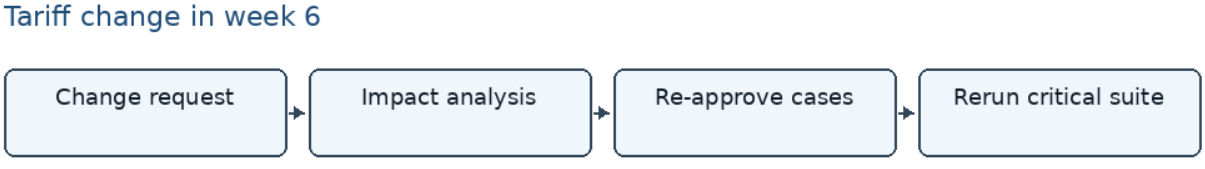


Figure: Tariff change in week 6 (illustrative process diagram).

Recommendation and limitation

Do not edit a tariff table in place without versioning. If the revised rule arrives late, freeze the previous evidence as belonging to the old baseline, create a new test baseline and rerun the impacted set. If time cannot support adequate validation, delay release or formally reduce scope only where safe.

TASK 2.9 Features excluded and suspension criteria

Prediction: exclusions should be explicit, justified and bounded; no critical billing/payment risk should be excluded solely to meet a calendar date.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Reasonable exclusion candidates	Third-party provider internals outside contract; unsupported legacy browsers; cosmetic content not linked to legal obligations.	Test interface contracts and supported browser list instead.
Not acceptable to exclude	500-unit boundary, duplicate payment, bill/payment consistency, access control, release-build identity, data masking.	These directly relate to pilot defects or legal obligations.
Suspend testing when	Build is unidentifiable, critical environment is unstable, test data is corrupted, or a critical blocker prevents reliable results.	Preserves validity of results.
Resume when	Correct build is tagged, environment smoke test passes, data reset is confirmed and owner approves.	Prevents untrusted runs being mixed with valid results.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Exclusion record	Feature, rationale, risk, compensating test and approver.	Makes residual risk visible.
Suspension evidence	Incident, affected cases, last valid run and environment status.	Avoids ambiguous blocked results.
Resume evidence	New build/environment ID and restart checklist.	Makes results comparable.

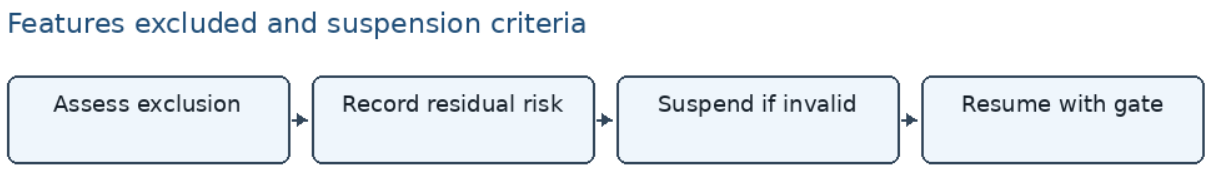


Figure: Features excluded and suspension criteria (illustrative process diagram).

Recommendation and limitation

Approve only exclusions that are documented, risk-reviewed and accepted by the proper authority. If the release build cannot be identified or the single server is unreliable, suspend affected tests instead of reporting false passes.

TASK 2.10 LLM-generated plan critique and final choice

Prediction: an LLM can draft a readable outline but may invent performance targets, omit approval roles or misunderstand cumulative slab tariffs. Its draft must be checked against the provided project facts.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication

Completeness review	Compare every section to IEEE-style checklist and assignment requirements.	Flag missing items rather than assuming they are implicit.
Fact check	Verify team size, ten-week schedule, 180 conditions, ₹1.5 lakh budget and six modules.	Reject fabricated people, tools, dates or service levels.
Domain review	Billing coordinator validates tariff, tax, rounding, meter corrections and boundary examples.	AI cannot approve commercial interpretation.
Security/audit review	Check privacy, evidence retention, build identity and approval rights.	Ensure legal and organisational constraints are present.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
LLM acceptance score	Score factual accuracy, completeness, specificity, traceability and actionability 0–2 each.	Require review and no critical omission.
Final plan choice	Complete plan with risk-based breakdown and iteration child plans.	Balances audit and execution.
Reviewers	Test manager, billing coordinator, build engineer, developer lead and Chief Engineer.	Shared ownership reduces single-person assumptions.

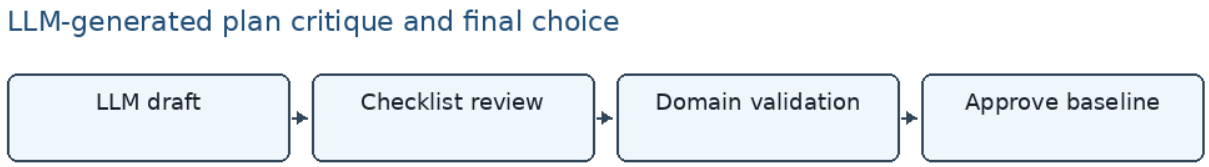


Figure: LLM-generated plan critique and final choice (illustrative process diagram).

Recommendation and limitation

Final decision: approve the complete master plan with risk-based test-level mapping and concise iteration plans. Revisit if scope, tariff policy or schedule changes. The process fails if the plan is treated as static paperwork; review it at each build baseline and material requirement change.

TASK 3.1 Three identification approaches

The pilot's tested-build/released-build mismatch means test identity must include more than a date or verbal statement. Prediction: baseline/configuration control combined with traceability will best prevent mismatch.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Build number	Lists build number and date.	Simple but can miss a changed tariff file or environment configuration under the same number.
B. Baseline/configuration items	Tags modules, tariff table, scripts, config and deployment manifest together.	Strong identity of the actual item set; requires disciplined release engineering.
C. Requirement traceability	Links requirements to design, cases, execution and build.	Shows coverage and change impact; does not alone prove deployed bytes match unless linked to baseline.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Detection test	Intentionally place an older tariff file in test environment.	Record detection time and evidence.

Identity evidence	Build ID + commit + tariff hash + config version.	Use one immutable baseline per test cycle.
Traceability	Requirement → design → case → execution → build.	Identify uncovered requirement and orphan test.

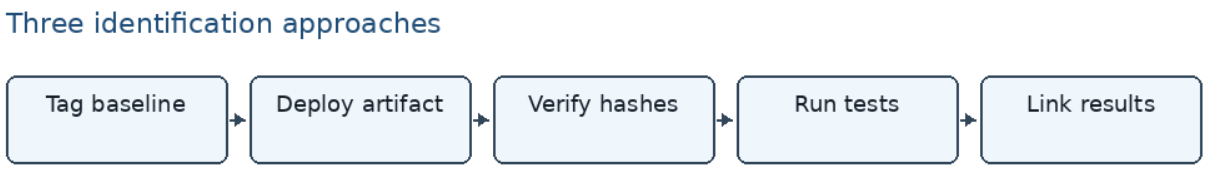


Figure: Three identification approaches (illustrative process diagram).

Recommendation and limitation

Make baseline/configuration-item identification compulsory and join it to requirement traceability. Build number alone is too weak for a portal where one tariff file can change the money charged. A traceability matrix without an actual deployed hash also cannot prove configuration identity.

TASK 3.2 Test item inventory

This inventory is the starting point for the plan and must distinguish included items from explicit exclusions.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Application items	Registration/login, meter upload, billing, payment, receipt/SMS, complaint/outage modules.	All six modules are in scope.
Configuration items	Tariff table, tax/fee settings, feature flags, endpoint configuration, SMS templates.	Version and hash each release-relevant item.
Testware	SRS, test plan, design spec, cases, procedures, synthetic data, scripts, logs and incident reports.	Testware itself needs version control.
Excluded items	Payment provider internal systems, unsupported browsers, real consumer data in unmasked form.	Use contract tests and approved support matrix.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Identifier format	PP-BLD-1.0.0+commit; TARIFF-2026-01-v3; ENV-STG-01.	Readable and unique.
Owner	Build engineer owns manifest; test manager verifies; module owner confirms version.	Clear accountability.
Transmittal	Item, version, hash, source, destination, sender, receiver and time.	Creates custody record.

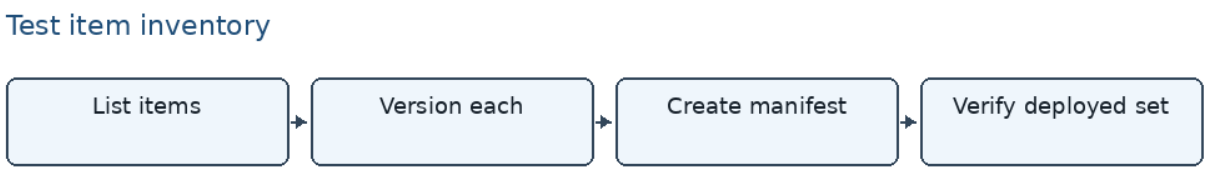


Figure: Test item inventory (illustrative process diagram).

Recommendation and limitation

Never write only 'latest build' in a test plan. Use a release manifest and compare it at test start, summary and release packaging. If any item differs, invalidate impacted results until assessed and rerun.

TASK 3.3 Test design specification: bill generation

This design specification defines the scope and approach for PP-R04 and PP-R05. Tariff semantics must be approved by the billing coordinator before expected values are final.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Feature	Slab-based electricity charge and displayed bill breakdown.	
Test approach	Equivalence partitioning, boundary value analysis, independent oracle calculation, negative input and regression.	
Test data	Synthetic consumer IDs; unit values around 100, 300 and 500; tariff version identified.	
Entry criteria	Approved tariff rules, tagged build, stable test server, test data reset and expected-value review.	
Exit criteria	All critical boundary cases pass; no open critical/high bill-total defect; result traceability complete.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Boundary set	99, 100, 101; 299, 300, 301; 499, 500, 501.	Covers adjacent values around slab boundaries.
Oracles	Independently calculate expected charge in a spreadsheet or reviewed helper.	Avoid testing code against itself.
Evidence	Input units, tariff version, slab breakdown, expected/actual total, build ID and result.	Supports reproduction and audit.

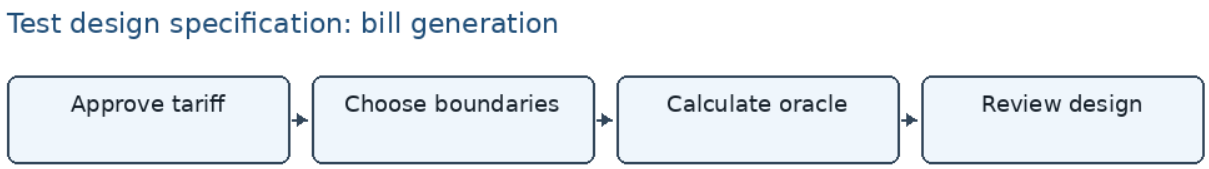


Figure: Test design specification: bill generation (illustrative process diagram).

Recommendation and limitation

Design should be reviewed by a tester and the billing coordinator. If the tariff applies marginal rates per band, expected charge is the sum of units in each band multiplied by that band's rate; if rates apply to all units at a threshold, the expected results differ materially. Do not proceed with ambiguous rules.

TASK 3.4 Four test case specifications

The four cases below are minimum representative cases. Extend them to cover all boundaries, invalid values, rounding and tax as defined by the approved SRS.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
TC-BILL-01	Input 100 units; expected slab boundary result according to approved tariff.	Valid lower boundary.
TC-BILL-02	Input 101 units; verify first unit of next slab is handled correctly.	Off-by-one check.
TC-BILL-03	Input exactly 500 units; verify approved boundary behaviour and displayed slab totals.	Historical critical fault.
TC-BILL-04	Input 501 units; verify above-boundary rate applies only as specified.	Adjacent boundary and regression.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Preconditions	Synthetic consumer active; meter reading accepted; tariff version tagged.	Repeatable setup.
Steps	Create reading → calculate bill → open bill breakdown → compare independent oracle.	Procedure should be executable by another tester.
Expected output	Correct total, correct slab quantities, no negative/duplicated units and correct tariff ID.	Record numeric expected values only after tariff approval.

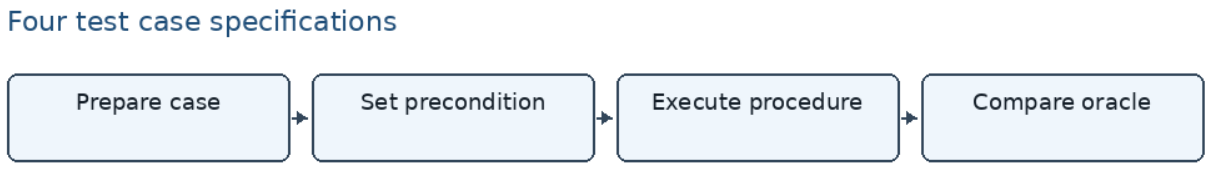


Figure: Four test case specifications (illustrative process diagram).

Recommendation and limitation

Keep expected amounts in the approved test data file and link the file version to the case. The report should not hardcode an assumed tariff interpretation as fact. For a tariff table defined as marginal bands 0–100 at ₹3, 101–300 at ₹5, 301–500 at ₹7 and above 500 at ₹9, clarify whether the band widths are 100, 200, 200 and unbounded respectively.

TASK 3.5 Test procedure and item transmittal

A test procedure describes execution order and recovery steps; the item transmittal report establishes what was handed to the test environment.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Procedure step 1	Confirm build ID, commit and tariff/configuration hash; reject mismatch.	Build engineer + tester.
Procedure step 2	Reset synthetic data and create meter reading with boundary units.	Tester.
Procedure step 3	Generate bill; save UI/API response, calculation breakdown and logs.	Tester.
Procedure step 4	Compare against independent expected result; file incident if mismatch.	Tester; billing coordinator validates rule.

Transmittal fields	Item ID/version/hash, source, destination, date/time, sender, receiver and known issues.	Build engineer owns record.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Repeatability	A second tester can run the same steps without oral explanation.	Target reproducibility rate is 100% for critical incidents.
Chain of custody	Manifest follows build from CI to test server to release candidate.	Avoids silent replacement.
Invalid transmittal	Any mismatch in hash/version blocks execution or triggers formal impact analysis.	No result may be attached to an unknown build.

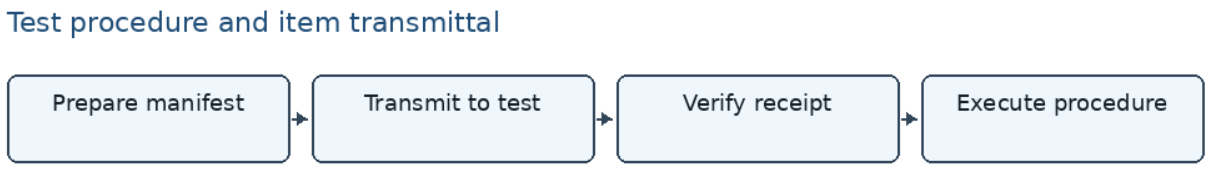


Figure: Test procedure and item transmittal (illustrative process diagram).

Recommendation and limitation

Make transmittal mandatory at every candidate deployment and release. A test procedure without environment and data setup is incomplete; a manifest without a deployment verification does not prove what actually ran.

TASK 3.6 Simulated wrong tariff file exercise

Prediction: baseline/hash verification will detect the old tariff file immediately; build number alone may not. This is a designed exercise, not a claimed live deployment.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Build-number only	Build ID remains the same while tariff file is replaced.	Likely miss unless file version is separately recorded.
B. Baseline/configuration	Compare deployed tariff hash and version with manifest.	Detects mismatch at environment smoke check.
C. Traceability	Cases reference tariff version and expected data baseline.	Detects mismatch when result is linked/validated; fastest if tool enforces version precondition.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Detection time	Start timer at wrong-file deployment; stop at alert or first human discovery.	Measure in minutes during actual simulation.
Runs invalidated	Count cases executed against wrong tariff version.	Mark results invalid and rerun after correction.
Lessons	Capture control that detected it and control that failed.	Avoid attributing detection to a control that was not active.

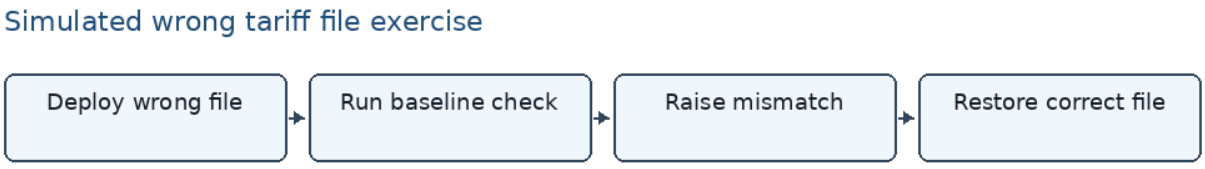


Figure: Simulated wrong tariff file exercise (illustrative process diagram).

Recommendation and limitation

Compulsory control: deployment pipeline validates tariff/configuration hashes before test execution; traceability tool stores the baseline identifier per result. If the check is skipped, the team may report a false pass on the wrong tariff and ship a financial defect.

TASK 3.7 Traceability matrix and orphan review

Traceability coverage is calculated against PP-R01 to PP-R14. Every requirement must link to at least one case or have an approved reason for exclusion; every case must link to a requirement or risk.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
PP-R01 to R03	Registration, login, meter-reading validation.	Functional + security tests.
PP-R04 to R06	Tariff calculation, bill breakdown and bill uniqueness.	Boundary + reconciliation tests.
PP-R07 to R10	Payment idempotency, reconciliation, receipt and SMS.	Integration + duplicate/timeout tests.
PP-R11 to R12	Complaint and outage reporting.	Workflow + notification tests.
PP-R13 to R14	Load and staff access/audit.	Performance + security/audit tests.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Requirement coverage	Requirements with linked cases ÷ 14 × 100%.	Target 100% mapped, including explicitly deferred requirements.
Test orphan rate	Cases without requirement/risk link ÷ total cases.	Target 0% unapproved orphans.
Change impact	Requirements and cases touched by a tariff revision.	R04/R05 and related payment/receipt cases likely impacted.

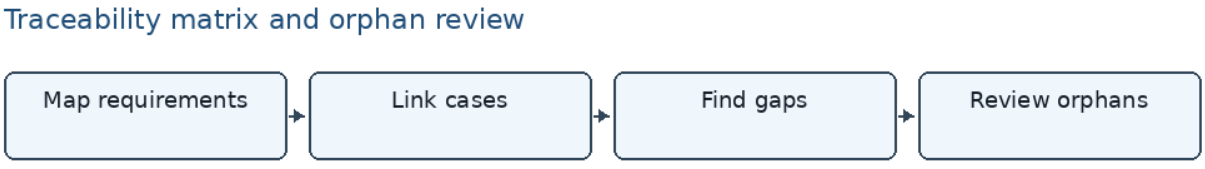


Figure: Traceability matrix and orphan review (illustrative process diagram).

Recommendation and limitation

Traceability is not just a spreadsheet count. Review whether the linked test actually verifies the requirement. An LLM may invent plausible links or miss negative paths; validate every link by reading the requirement, test precondition, action and expected result.

TASK 3.8 Task 3 final recommendation and LLM check

The final attachment set should be small enough to use and strong enough to reproduce the bill boundary test and prove the tested artifact.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Mandatory attachments	SRS excerpt, test design specification, four or more case specs, test procedure, item transmittal report, traceability matrix and baseline manifest.	Directly address pilot failure modes.
Lean supporting records	Tool-generated execution logs and summary metrics.	Avoid duplicating case data in Word.
LLM matrix generation	Can propose requirement-case links from stable IDs.	May hallucinate coverage or overlook security/performance cases.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Review method	Sample each requirement and inspect linked expected result; inspect every critical requirement.	Human reviewer signs.
Completeness target	100% critical requirement coverage; all 14 requirements mapped or explicitly deferred.	No hidden gaps.
Documentation cost	Estimate authoring/maintenance hours and compare with avoided rework.	Use templates and automated export where safe.

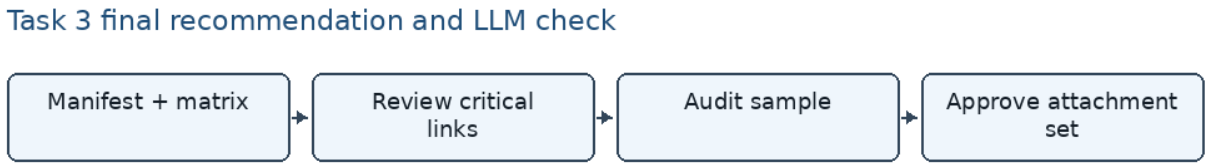


Figure: Task 3 final recommendation and LLM check (illustrative process diagram).

Recommendation and limitation

Compulsory choice: tagged baseline plus requirement traceability. A build number is necessary but insufficient. The approach fails if hashes are not verified on the deployed environment or if people update the matrix without updating tests. Tool automation should reduce clerical effort, not replace review.

TASK 4.1 Reporting approaches and audience

Prediction: a dashboard plus incident reports and level summary will answer the Chief Engineer's readiness question better than any single reporting format.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Logs and incident reports only	Strong detail and chronology; no roll-up or readiness decision.	Developers can reproduce issues but leadership cannot see total progress.
B. Daily dashboard + weekly status	Shows executed/passed/failed/blocked, defects by severity, risks and actions.	Fast operational control; percentages can mislead without severity and scope.
C. Test summary report	Evaluates comprehensiveness, deviations, results, residual risk and approvals.	Best formal gate; arrives too late if used without daily reporting.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Chief Engineer	Needs readiness, critical risk, schedule and decision request.	Dashboard plus summary.
Developer	Needs exact reproduction and evidence.	Incident report and log.
Auditor	Needs scope, baseline, coverage and approvals.	Traceable summary and archived execution records.

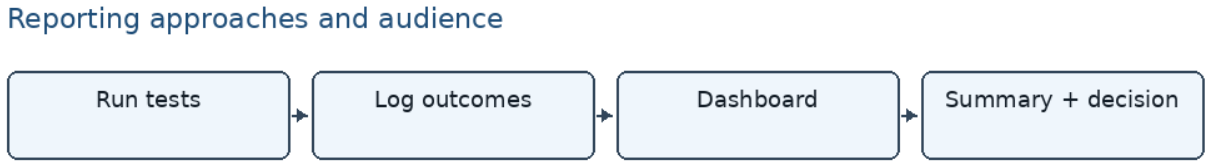


Figure: Reporting approaches and audience (illustrative process diagram).

Recommendation and limitation

Use B during execution, A for every failure and C at each test-level exit. A 95% pass rate is not readiness if the remaining 5% includes a critical billing or duplicate-payment defect.

TASK 4.2 Simulated execution set: 15 cases

The table is a simulated training dataset to demonstrate reporting structure. Replace statuses with real results if the assignment requires actual execution. A test log records every attempt, not only failures.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
TC-01 Login valid	PASS	Build PP-0.8.1 / synthetic user
TC-02 Invalid password	PASS	Access denied and event logged
TC-03 Meter reading valid	PASS	Reading accepted
TC-04 Negative units	PASS	Rejected with validation message
TC-05 Bill at 100 units	PASS	Matches approved oracle
TC-06 Bill at 101 units	PASS	Next-slab boundary checked
TC-07 Bill at 500 units	FAIL	Critical incident IR-01
TC-08 Bill at 501 units	PASS	Adjacent boundary checked
TC-09 Bill breakdown total	PASS	Components sum correctly
TC-10 Payment success	PASS	Sandbox response confirmed
TC-11 Duplicate payment retry	FAIL	Critical incident IR-02
TC-12 Gateway timeout	PASS	Pending then reconciled
TC-13 Receipt only after success	PASS	No receipt on declined payment
TC-14 SMS receipt	BLOCKED	Sandbox SMS service unavailable
TC-15 Complaint creation	PASS	Complaint ID generated

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Executed	13 completed outcomes out of 15 planned; 1 blocked and 1?	Count must be recalculated from the actual status list.
Failures	2 failures in the illustrative list.	Both include critical financial risks.
Blocked	SMS case blocked by dependency.	Not a pass; retest required.

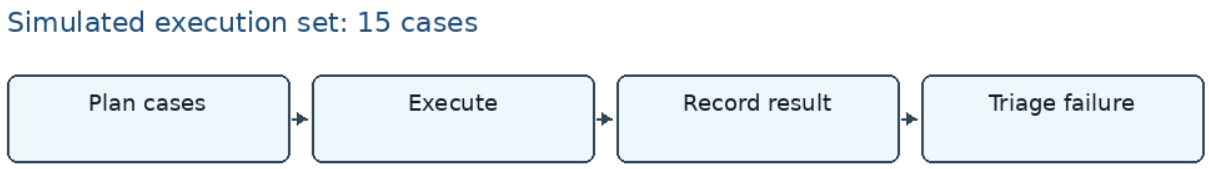


Figure: Simulated execution set: 15 cases (illustrative process diagram).

Recommendation and limitation

Important count check: the displayed sample contains 15 cases, 12 PASS, 2 FAIL and 1 BLOCKED. Report raw counts alongside percentages and name the critical failures. Do not call blocked cases passed or remove them from the denominator without stating the rule.

TASK 4.3 Test log template and example records

A useful log contains case ID, build, environment, tester, start/end, input, result, evidence link and incident ID. It must allow the run to be repeated.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Log field	Example value	Purpose
Run ID	RUN-2026-11-04-01	Unique execution batch.
Build / baseline	PP-0.8.1 / TARIFF-v3 hash verified	Defines tested artifact.
Environment	STG-01, payment sandbox	Allows comparison.
Test case / input	TC-07 / 500 units	Reproduces boundary.
Expected / actual	Approved tariff total / displayed total differs	Shows the failure.
Status / incident	FAIL / IR-01	Connects test and defect.
Timestamp / tester	Recorded by assigned tester	Audit trail.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Log quality	All fields populated for critical tests.	No orphan result.
Run reproducibility	Second tester follows steps and gets same outcome.	Record success/failure.
Retention	Keep raw export and evidence references with access control.	Supports audit.

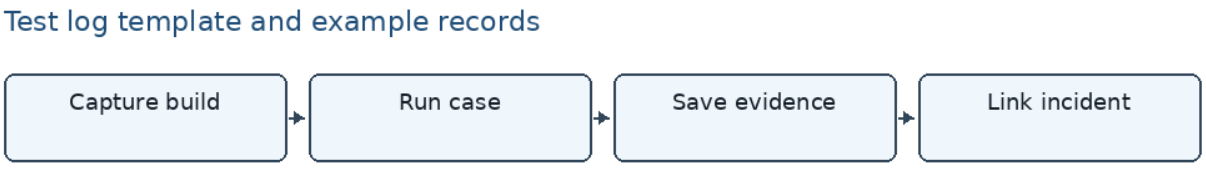


Figure: Test log template and example records (illustrative process diagram).

Recommendation and limitation

Logs should be append-only or change-tracked. Correct a mistaken result by adding a superseding record rather than silently overwriting history. Separate test execution failure from environment failure and product defect.

TASK 4.4 Incident report IR-01: bill at 500 units

Incident template: unique ID, summary, environment, preconditions, steps, expected/actual result, severity, evidence, reproducibility and related requirement.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Summary	Bill total differs at exactly 500 units; suspected off-by-one slab boundary.	
Severity	Critical until financial impact is bounded; billing coordinator confirms impact.	
Steps	Use synthetic consumer; set units=500; generate bill; compare slab breakdown with approved tariff version.	
Expected / actual	Expected total from independent oracle; actual portal total differs (numeric values to be entered from run).	
Evidence	Screenshot/API response, tariff hash, build ID, calculation log and test data ID.	
Owner / next action	Billing developer investigates; tester reruns 499/500/501 after fix.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Reproducibility	Another tester receives only this report and follows the steps.	Target: reproduce without oral hints.
Impact	Potentially affects consumers at a boundary; quantify affected readings after root cause.	Avoid assuming all bills affected.
Closure	Fix commit plus failing-before/passing-after regression.	No closure on developer assertion alone.

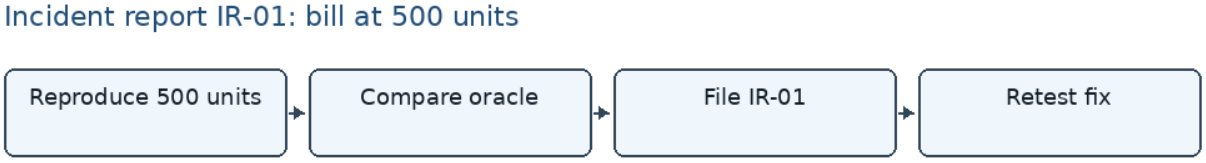


Figure: Incident report IR-01: bill at 500 units (illustrative process diagram).

Recommendation and limitation

Do not invent a numeric expected bill amount until the tariff interpretation is approved. The incident must attach the approved tariff version and independent calculation. If the report omits the build or units, reproduction will be unreliable.

TASK 4.5 Incident report IR-02: duplicate payment

This incident represents the second critical pilot-class risk. Payment retry and delayed callbacks must not create a second charge or duplicate ledger entry.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Summary	Repeated payment request/callback creates two records for one bill.	
Preconditions	Synthetic bill; sandbox payment instrument; unique transaction reference.	
Steps	Submit payment; simulate timeout or duplicate callback; retry using same idempotency key; inspect ledger and receipt list.	
Expected	Exactly one financial charge and one receipt; state reconciles to gateway.	
Actual	Two records appear in the simulated failure case; attach ledger export and gateway event IDs.	
Severity / owner	Critical; payment developer + test manager; release blocked until verified.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Idempotency	Same key repeated produces one charge.	Required for retries and callbacks.
Recovery	Timeout followed by success must reconcile without second charge.	Test asynchronous order.
Closure	Replay duplicate event after fix and inspect ledger.	Regression evidence retained.

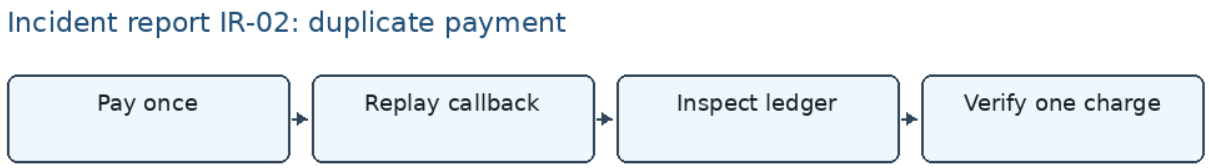


Figure: Incident report IR-02: duplicate payment (illustrative process diagram).

Recommendation and limitation

Reproducibility depends on a controlled sandbox and deterministic event replay. Never use real bank credentials or real consumer data in the assignment artefact. A duplicate ledger record is a critical defect even if the UI shows only one receipt.

TASK 4.6 Incident report IR-03: environment / receipt blocker

The third incident should represent a distinct class such as SMS receipt blocked by an unavailable sandbox. A blocked test is not a product failure until dependency status is established.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Summary	SMS receipt case cannot complete because test SMS gateway is unavailable.	
Classification	Blocked/environment dependency, pending confirmation; do not classify as PASS.	
Steps	Complete sandbox payment; invoke receipt workflow; capture gateway availability response and correlation ID.	
Expected	One SMS receipt queued/sent after confirmed payment.	
Actual	No delivery acknowledgement because the sandbox is unavailable.	
Action	Build engineer restores sandbox; rerun; if gateway is healthy and portal still fails, raise product incident.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Evidence sufficiency	Logs, timestamps, endpoint status and correlation ID.	Separates environment issue from code defect.
Reproducibility	Second tester repeats after service restoration.	Confirm outcome.
Reporting	Count as blocked until rerun.	Keeps pass rate honest.

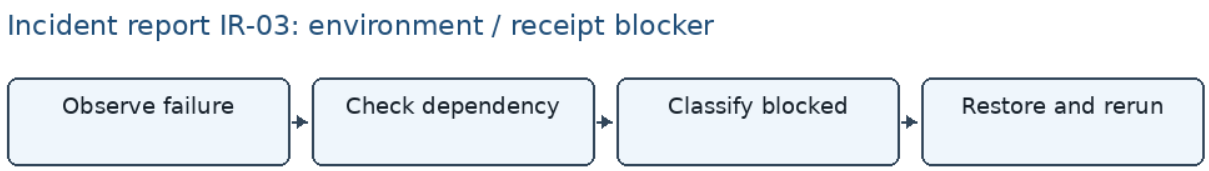


Figure: Incident report IR-03: environment / receipt blocker (illustrative process diagram).

Recommendation and limitation

Incident reports must distinguish severity, priority and type. The report should not assign a product defect merely because a third-party test dependency is down; equally, teams must not hide repeated environment blockers because they reduce reported progress.

TASK 4.7 Daily dashboard and weekly status report

A dashboard should show scope and risk, not only a percentage. The following values are illustrative and should be recalculated from the final log.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Metric	Illustrative value	Interpretation
Planned / completed	15 planned / 14 attempted, 1 blocked	Clarify whether blocked is attempted or not executed.
Status counts	12 PASS / 2 FAIL / 1 BLOCKED	Counts sum to 15.
Defects by severity	2 critical open, 0 high, 0 medium	Two criticals block release.
Coverage	15 of 180 functional conditions executed in sample	8.3% of condition inventory; sample is not readiness evidence.

Top risks	500-unit bill; duplicate payment; SMS sandbox	Owners and due dates required.
Decision request	Hold candidate until critical defects fixed and blocked receipt test rerun.	Actionable recommendation.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Weekly narrative	Progress, variance, next-week plan, blockers, decisions needed.	Explains why metrics moved.
Trend	New/closed/reopened defects and execution velocity by day.	Reopen rate and age matter.
Readiness	Critical criteria met? Traceability complete? Candidate build identified?	A green pass percentage cannot override critical failure.

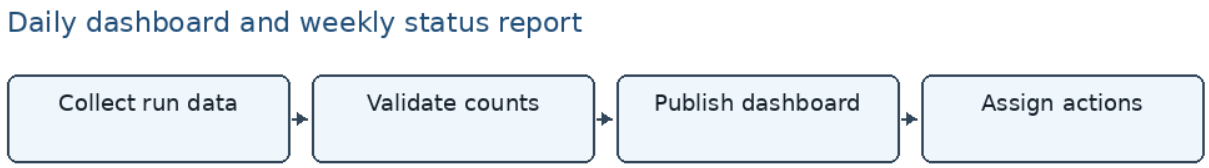


Figure: Daily dashboard and weekly status report (illustrative process diagram).

Recommendation and limitation

Send a concise dashboard daily and a narrative status report weekly. Include denominator, severity, baseline and blocked count. If the dashboard is generated automatically, the test manager still validates definitions and data quality.

TASK 4.8 Test summary report and reporting decision

The summary report evaluates the test level against the plan. It should document deviations, comprehensiveness, result summary, unresolved risks and release recommendation.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Summary fields	Test item/build, scope, environment, executed cases, pass/fail/blocked, incidents, coverage, deviations and approvals.	
Comprehensiveness	Requirement and risk coverage; known untested areas; reasons for exclusions.	
Evaluation	Compare exit criteria with evidence, not opinion.	
Recommendation	Release / hold / conditional only where policy allows and risk is accepted by authorised owner.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Effort	Log: ongoing as executed; dashboard: small daily refresh; summary: planned review and sign-off.	Estimate actual hours in time log.
Reproducibility audit	Give three incidents to classmate without verbal guidance.	Record number reproduced and missing detail.
LLM summary critique	Compare every claim to raw log; reject unsupported readiness claims.	No invented metrics or omitted critical failures.

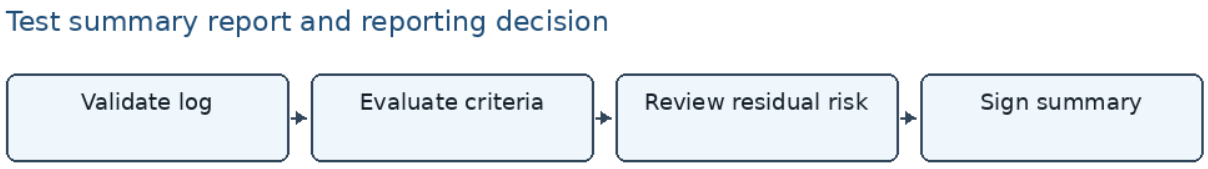


Figure: Test summary report and reporting decision (illustrative process diagram).

Recommendation and limitation

Recommended set: logs and incidents for every run/failure, daily dashboard and weekly status during execution, and formal summary at each level/release gate. The approach fails if metrics are generated from stale exports or a critical failure is hidden by aggregate pass percentage.

TASK 5.1 Three policy approaches

Prediction: a formal policy with a strategy and improvement path will be more measurable than informal guidelines, but adoption depends on simplicity and ownership.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Informal guidelines	One-page practices are easy to distribute and follow; ambiguous and weak for audit or release authority.	Good starter for a tiny feedback form; insufficient for PowerPay.
B. Formal organisational policy	Signed mission, principles, levels, standards, documents, metrics, targets, roles and release rules.	Enforceable and auditable; needs governance and review.
C. Policy + strategy + improvement path	Formal policy plus test strategy, maturity improvement roadmap and measurement cycle.	Best long-term approach; may be excessive if launched as bureaucracy without coaching.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Pilot prevention	Map each policy rule to wrong bill, duplicate payment, wrong build and missing test evidence.	A rule only helps if evidence is checked.
Adoption likelihood	Estimate ease, training burden and enforcement.	Policy compliance is operational, not merely signed.
Cost	Drafting, workshops, tools, training and audit effort.	Budget should include ongoing maintenance.

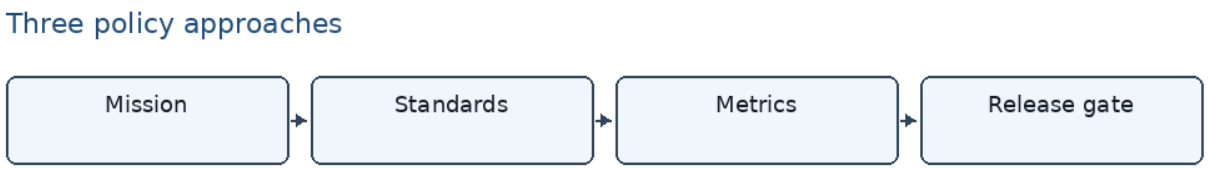


Figure: Three policy approaches (illustrative process diagram).

Recommendation and limitation

Adopt B as the enforceable minimum and C as the improvement roadmap, rolled out in stages. Informal guidance remains a quick-reference summary, not the authoritative policy.

TASK 5.2 Policy A: informal guidelines

The one-page version is intentionally simple. It should point to authoritative templates and not attempt to replace formal approval.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Scope	All portal projects and maintenance changes; tailor depth to risk.	
Objectives	Find defects early, protect consumer money/data and provide reliable evidence.	
Roles	Developers test unit logic; testers independently design/exercise tests; business owner validates rules; release authority approves.	
Release rule	Do not release with unresolved critical billing, payment, privacy or build-identity defects.	
Documentation	Record test cases, results, incidents and a brief summary.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Strength	Low drafting cost and quick onboarding.	High initial adoption potential.
Weakness	No formal exception process or detailed metric definitions.	Teams may interpret 'tested enough' differently.
Pilot risk	Could encourage local fixes without a regression test or build proof.	Wrong-build problem remains possible unless explicitly stated.

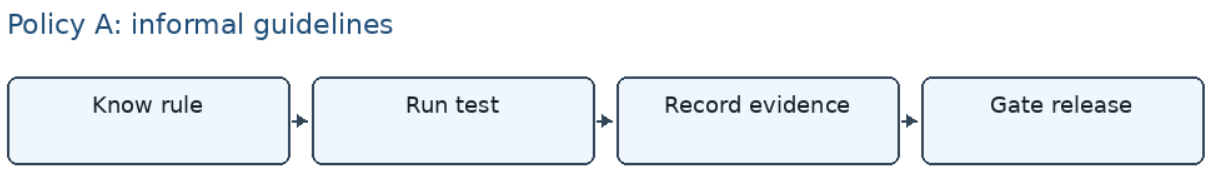


Figure: Policy A: informal guidelines (illustrative process diagram).

Recommendation and limitation

Use as a laminated quick reference and induction sheet. It fails if staff treat recommendations as optional or if the organisation cannot audit compliance. A short rule still needs a defined owner and measurable evidence.

TASK 5.3 Policy B: formal policy draft

This policy is the Corporation's normative baseline. Detailed project plans may tailor methods, but cannot waive critical integrity requirements without authorised governance.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Mission and principles	Testing reveals failures; independent expected values; early testing; risk prioritisation; traceable evidence; defects are not hidden by aggregate metrics.	
Levels	Unit, integration, system, regression, security/performance and user acceptance as risk requires.	

Documentation	Plan, test design/cases/procedure, log, incident, summary, build manifest and traceability.	
Metrics	Requirement coverage, executed status counts, critical defects, reopen rate, build identity compliance, defect age and performance results.	
Responsibilities	Test manager owns plan/report; build engineer owns artifact identity; billing owner approves tariff semantics; release authority signs readiness.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Policy owner	Head of IT Quality; annual review and after major incident.	
Compliance evidence	Signed plan, tool export, audit sample and release decision.	
Exception	Document risk, compensating control, approver, expiry date; no exception for unknown build identity.	

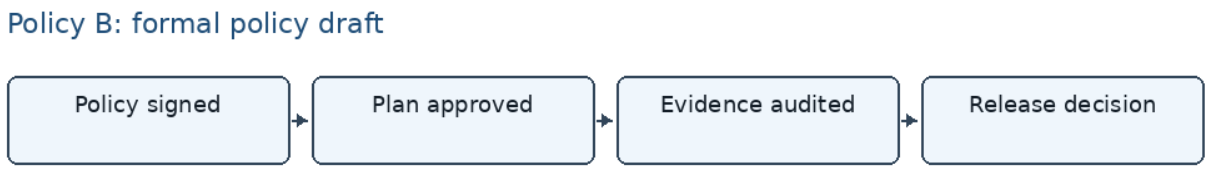


Figure: Policy B: formal policy draft (illustrative process diagram).

Recommendation and limitation

Formal policy addresses the pilot problems directly. Its weakness is that teams may create paperwork for every trivial change; tailor depth by risk while keeping mandatory controls for financial integrity, privacy and release identity.

TASK 5.4 Policy C: improvement roadmap

The improvement path should be staged so that teams can comply before maturity targets are raised. A maturity model is a guide, not a substitute for measured outcomes.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Stage 1 – repeatable	Common plan, incident template, build manifest and release gate; train all roles.	
Stage 2 – managed	Metrics, risk-based estimation, defect trend review, traceability and tool-supported evidence.	
Stage 3 – defined	Organisation-wide test strategy, standard templates, role competencies and periodic process audit.	
Stage 4 – optimisation target	Use trend data to improve test selection, automation and defect prevention after reliable baseline data exists.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Measure	Baseline first, then target and owner.	Avoid claiming maturity based only on documentation.
Improvement cadence	Monthly quality review; quarterly policy audit.	Actions have due dates.

Failure condition	Too many forms, no tool integration, no team feedback.	Simplify low-risk documents and automate repetitive capture.

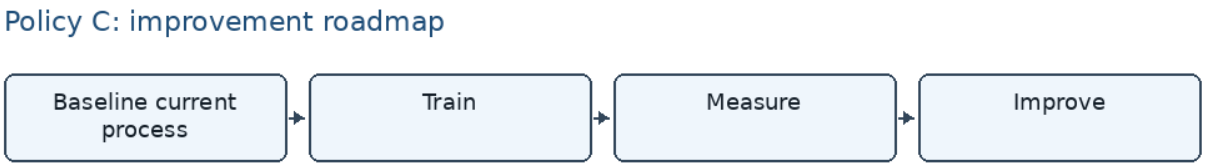


Figure: Policy C: improvement roadmap (illustrative process diagram).

Recommendation and limitation

Use C as the policy's improvement plan, not a promise that a maturity level has been achieved. Improvement is justified by evidence such as lower reopen rate, better reproducibility and no tested/released baseline mismatch.

TASK 5.5 Measurable policy statements

Policy statements should have clear definitions and data sources. Targets are proposed for the exercise and require approval by the Corporation.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
P-01 Critical defects	Zero open critical billing/payment/privacy defects at release.	Measure from defect tool at release gate.
P-02 Build identity	100% of test runs linked to immutable build and configuration baseline.	Audit random execution records.
P-03 Critical regression	Every payment and tariff change has automated or repeatable regression cases.	Review change-to-test traceability.
P-04 Requirement coverage	100% critical requirements mapped to tests; all remaining gaps justified.	Traceability export.
P-05 Incident quality	At least 90% of sampled incidents reproducible from report alone; 100% for criticals.	Independent reproduction exercise.
P-06 Reopen rate	Reduce from pilot baseline of 18% over a defined rolling period.	Reopened defects / defects marked fixed; report denominator and period.
P-07 Test evidence	100% critical results include timestamp, build, environment and evidence reference.	Audit sample.
P-08 Privacy	Zero unmasked personal consumer records in test artefacts/logs.	Automated scan and manual audit.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Metric owner	Assign owner for extraction and review.	Prevents unowned dashboard fields.
Target caveat	Targets need a measurement window and exclusions definition.	Avoid comparing unlike datasets.
Escalation	Critical breach blocks release and triggers incident review.	Enforcement makes policy real.

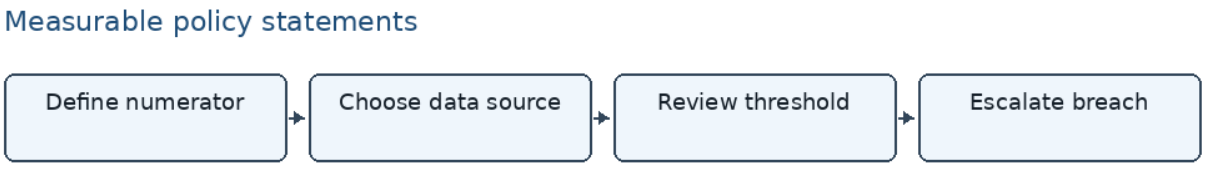


Figure: Measurable policy statements (illustrative process diagram).

Recommendation and limitation

Use these targets as proposed policy controls. A metric is not measurable unless numerator, denominator, period, data source and owner are defined. Do not promise a reduction in reopen rate without recording a baseline and follow-up period.

TASK 5.6 Policy versus pilot failure matrix

This matrix connects each pilot-year failure to the policy controls that would prevent or detect it.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Wrong bill at 500 units	Boundary-value policy, independent oracle, billing-owner review and mandatory regression.	Prevents ambiguous expected values and catches off-by-one.
Duplicate payment	Idempotency test policy, payment integration cases, critical release gate and reconciliation evidence.	Detects repeated request/callback defects.
Wrong build released	Immutable baseline, deployment hash verification and release manifest sign-off.	Prevents unverified artifacts.
Unknown testing amount	Mandatory execution logs, dashboard and summary report with denominator.	Provides evidence of coverage and gaps.
18% reopen rate	Fix verification, regression attachment and defect root-cause review.	Reduces weak closure practice.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Small feedback-form project	Use short plan, essential cases, privacy check and brief summary.	Same principles, lower document volume.
PowerPay project	Full risk-tailored plan, traceability, test levels, payment/billing gates and formal sign-off.	Higher consequence warrants deeper evidence.
Tailoring boundary	Cannot remove build identity, privacy and critical financial checks.	Low-risk tailoring is not permission to omit core safeguards.

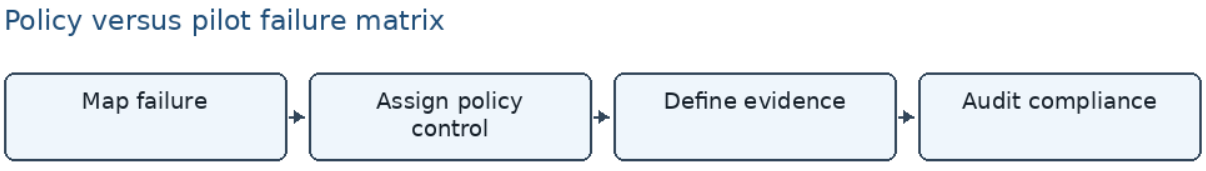


Figure: Policy versus pilot failure matrix (illustrative process diagram).

Recommendation and limitation

Adopt a risk-tailored implementation of the formal policy. If the organisation fails to enforce it, a signed policy becomes decorative. Audit real release decisions and report exceptions, not just the existence of documents.

TASK 5.7 Cost, effort and adoption

Costs below are planning estimates, not vendor quotes. The ₹1,50,000 tool budget must cover licence/configuration/training where applicable; free tools still require staff effort.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Informal guidelines	Draft/review: 8–16 staff-hours; minimal tool cost.	Easy to adopt but weak enforcement.
Formal policy	Workshops, drafting, legal/privacy review and approval: 40–70 staff-hours.	Moderate setup; stronger governance.
Policy + improvement	Formal policy plus baseline assessment, training and quarterly reviews: 80–140 initial staff-hours.	Highest cost; sustainable if leadership sponsors it.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Tool spend	Prefer existing issue tracker, test management and version control; budget paid licences only after evaluation.	Keep within ₹1,50,000.
Adoption measure	Training completion, policy checklist use, audit pass rate and exception rate.	Measure behaviour, not acknowledgements.
Review trigger	Major incident, new privacy law, major architecture change or annual cycle.	Policy remains current.

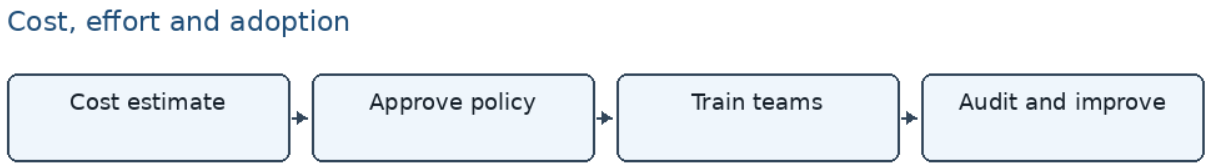


Figure: Cost, effort and adoption (illustrative process diagram).

Recommendation and limitation

Recommended rollout: issue a concise quick guide immediately, approve the formal policy before test execution, then run the improvement path over subsequent projects. If resources are limited, do not defer critical build and payment controls while waiting for a maturity programme.

TASK 5.8 LLM policy review and final choice

AI can draft policy language but may insert inappropriate standards, impossible targets or unclear authority. Human owners must align it with Corporation governance and local law.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Specificity	Names PowerPay risks, build identity, tariff boundary, idempotency, privacy and evidence.	Generic 'test thoroughly' statements are inadequate.
Measurability	Each rule has a threshold, owner, period and source.	Vague goals cannot be audited.
Consistency	Definitions of critical severity, release gate and exception authority agree across documents.	Contradictions create loopholes.
Legal review	Data masking, access and retention controls reviewed by authorised privacy/legal role.	AI is not legal approval.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Score	Specificity, measurability, consistency, completeness and usability scored 0–2.	Reject any critical gap.
Final selection	Formal policy plus staged improvement path and quick guide.	Cost balanced with audit need.
Failure if omitted	Teams follow local custom and cannot demonstrate consistent readiness.	Repeat pilot failures remain likely.

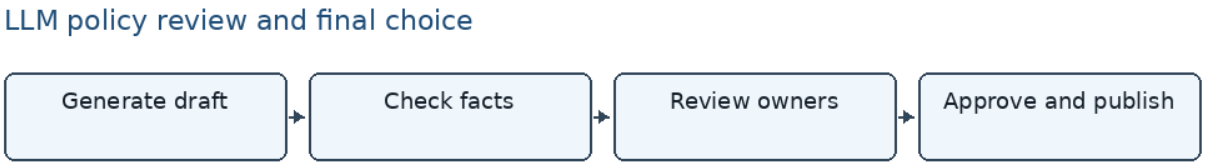


Figure: LLM policy review and final choice (illustrative process diagram).

Recommendation and limitation

Final choice: B is the enforceable policy, supported by C's improvement roadmap and A's quick-reference page. Reassess if policy adoption becomes paperwork-heavy; automate evidence and remove duplicative low-risk narrative while retaining mandatory controls.

TASK 6.1 Team structures and resource map

Prediction: module-aligned cross-functional teams will preserve domain knowledge and speed feedback, provided the test manager keeps an independent release gate.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Test-level functional teams	Separate unit/integration, system and acceptance teams; clear specialist ownership.	More handoffs and duplicated context; scheduling overhead.
B. Module-aligned cross-functional	Developer(s), tester and analyst work together for billing, payment and consumer service.	Fast local decisions and domain knowledge; independence must be protected.
C. Vendor system-test team + internal acceptance	Vendor handles system/performance; internal team owns requirements and acceptance.	Scalable specialist capacity but contract coordination and knowledge-transfer risk.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
People available	1 manager + 4 testers + 5 developers + 1 build engineer + 1 BA + 1 billing coordinator = 13.	No additional people assumed.
Interfaces	Test manager reports status to Chief Engineer; developers fix; build engineer controls baselines; billing coordinator validates tariff.	Clear escalation path.
Independence	Tester can report failures without developer veto; release authority remains internal.	Avoid conflict of interest.

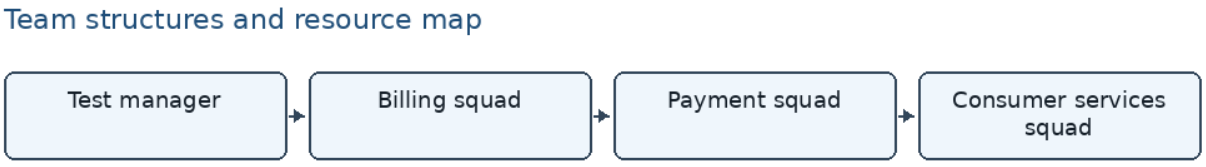


Figure: Team structures and resource map (illustrative process diagram).

Recommendation and limitation

Choose B with central test management and an independent release gate. Keep vendor support as a contingency for specialised performance testing if the internal environment or skills are insufficient; do not give a vendor unilateral release approval.

TASK 6.2 Organisation chart: functional test teams

A functional arrangement can separate responsibilities but creates hand-offs between test levels. The chart should name interfaces to development and business roles.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Unit/integration lead	Coordinates component checks with five developers; owns interface smoke tests.	
System test lead	Coordinates end-to-end, performance and security checks on the shared server.	
Acceptance lead	Coordinates billing coordinator, BA and consumer representatives.	
Build engineer	Deploys immutable candidates, manages environment schedule and transmittal.	
Test manager	Resolves priority conflicts and approves test evidence completeness.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Communication paths	Testers → test-level lead → test manager; developers and billing coordinator join defect triage.	Clear but more hops.
Single point of failure	Each level lead can become a bottleneck.	Cross-train and appoint deputies.
Delay risk	A defect discovered at system level may return through several teams.	Use common defect record and daily triage.

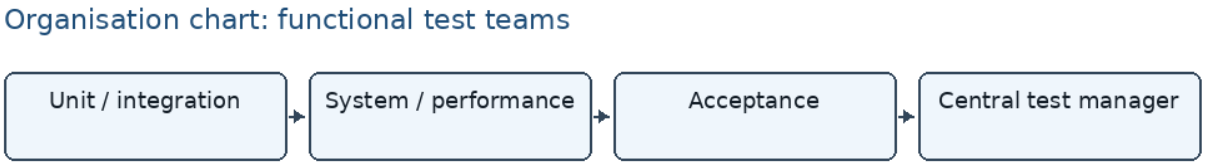


Figure: Organisation chart: functional test teams (illustrative process diagram).

Recommendation and limitation

Use functional teams where specialist test-level expertise is needed, but keep one shared plan, defect taxonomy and baseline manifest. This structure fails if hand-offs become queues and module knowledge is lost between test levels.

TASK 6.3 Organisation chart: module-aligned teams

Each module team owns design-to-regression work while shared controls remain central. The four testers can be allocated with one tester floating for risk and absence cover.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Billing module	2 testers (one billing-domain lead), 2 developers, BA support and billing coordinator for tariff approval.	
Payment/receipt	1 tester, 2 developers, payment interface owner; test manager reviews critical results.	
Consumer services	1 tester, 1 developer for registration/login, complaint/outage and SMS workflow; shared developer support as needed.	
Shared engineering	Remaining developers support integration and defects; build engineer owns deployment; test manager owns plan/report.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Urgent tariff change	Billing tester and coordinator impact-analyse immediately; developer branches fix; test manager reprioritises regression.	Fast local decision with formal approval.
Tester absence	Cross-trained floating tester covers critical billing/payment cases from procedures.	Lower key-person risk.
Dispute	Test evidence and requirement rule go to neutral triage; release gate is not controlled by disputed developer.	Maintains independence.

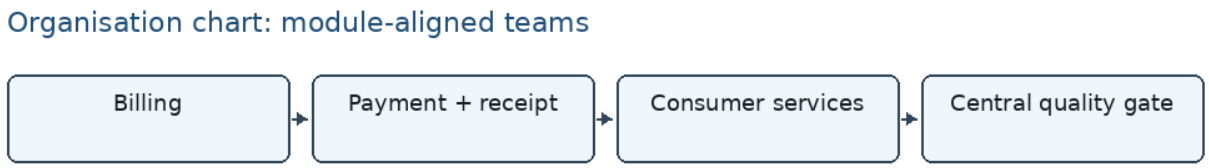


Figure: Organisation chart: module-aligned teams (illustrative process diagram).

Recommendation and limitation

Recommended structure: module-aligned teams with central standards, daily cross-team triage and independent sign-off. It supports the ten-week window by reducing hand-offs while retaining auditable control.

TASK 6.4 Organisation chart: vendor system testing

A vendor can bring performance and independent system-test capability, but internal staff must own acceptance and understand the results.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Vendor scope	System/performance test design and execution under agreed workload and evidence requirements.	
Internal test team	Requirements traceability, billing/payment critical regression, acceptance and defect triage.	

Contract controls	Environment, data masking, evidence format, defect severity, turnaround and ownership defined in statement of work.	
Release authority	Chief Engineer / authorised internal board; vendor recommends only.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Benefit	Specialist load tooling and independent perspective.	Useful if internal skill gaps exist.
Risk	Vendor may not understand tariff semantics; evidence may not align with internal build.	Require baseline access, review and knowledge transfer.
Release veto	Vendor must not independently approve release.	Commercial accountability stays with Corporation.



Figure: Organisation chart: vendor system testing (illustrative process diagram).

Recommendation and limitation

Use a vendor only for bounded specialist tasks and retain internal ownership of billing correctness and release decisions. This structure fails when contracts measure test volume instead of risk coverage or when vendor results cannot be reproduced.

TASK 6.5 Situation simulation and delay comparison

The table is a structured prediction to be populated with actual role-play times. Do not report predicted minutes as observed simulation results.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Defect dispute	Functional teams: lead escalates; module teams: requirement and test evidence triage; vendor: contract escalation.	Measure minutes to decision and record escalation steps.
Urgent tariff change	Functional: change board and test-level replan; module: billing squad impact analysis; vendor: change request may require commercial approval.	Measure time to identify impacted cases.
Tester absent 1 week	Functional: another level team may cover after handover; module: cross-trained tester covers; vendor: staffing clause or internal substitute.	Measure uncovered critical tests and reassignment delay.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Decision time	Start at event announcement; stop when owner and action agreed.	Use same scenarios across structures.
Continuity	Count critical cases with named backup.	Target 100% critical coverage ownership.
Overhead	Count hand-offs and meetings for each situation.	Lower is not always better if independence is lost.

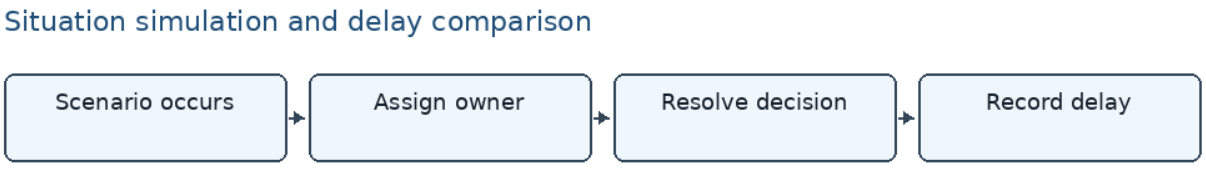


Figure: Situation simulation and delay comparison (illustrative process diagram).

Recommendation and limitation

Expected result: module-aligned teams should be fastest for tariff change and developer-tester collaboration; functional teams may provide stronger specialisation; vendor structure may scale performance work but has the highest coordination risk. Validate with the role-play and revise if evidence disagrees.

TASK 6.6 Final team recommendation and vendor critique

The team needs role clarity, skills coverage and a safe path for disagreement. Organisation design is successful only if it reduces risk without hiding defects.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Structure	Module-aligned cross-functional teams with central test manager and independent release board.	
Decision rights	Billing coordinator approves tariff meaning; developer owns fix; tester owns evidence; test manager owns status; Chief Engineer/release board owns go/no-go.	
Escalation	Critical financial/security issue → immediate test manager + Chief Engineer; no local override.	
Coverage	Backup owner named for every critical case and shared server slot.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Communication	Daily 15-minute defect/risk triage; weekly formal status.	Fast decisions with durable records.
Independence	Tester can raise incident directly in tool and escalate unresolved severity.	No pressure to downgrade for schedule.
Tool support	Shared repository, issue tracker, test management and build manifest.	One source of truth.

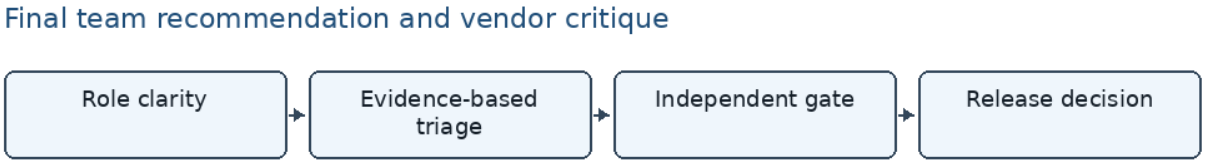


Figure: Final team recommendation and vendor critique (illustrative process diagram).

Recommendation and limitation

Do not grant the vendor the right to approve release. The vendor may attest that contracted tests passed, but the Corporation owns consumer money, legal obligations and residual risk. Reassess the team if cross-functional ownership causes a conflict of interest or if the shared server becomes the dominant bottleneck.

TASK 7.1 Skills matrix for four testers

The team profile is given at a high level: two fresh graduates, one billing-operations specialist and one former developer. Numeric scores below are proposed initial estimates and should be confirmed through assessment.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Technical	Test design, API basics, SQL/logs, automation, version control.	
Domain	Tariff slabs, meter readings, billing exceptions, payment reconciliation.	
Analytical	Boundary analysis, independent oracle, risk ranking, root-cause evidence.	
Communication	Clear incidents, triage, stakeholder explanation.	
Planning	Estimation, traceability, test data, reporting.	
Tools	Issue tracker, test management, CI logs, API client and spreadsheet.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Fresh graduate 1	Technical 2/5; domain 1/5; analytical 2/5; communication 3/5; planning 2/5; tools 2/5.	Needs fundamentals and guided practice.
Fresh graduate 2	Technical 2/5; domain 1/5; analytical 3/5; communication 2/5; planning 2/5; tools 2/5.	Needs report writing and test design.
Billing specialist	Technical 2/5; domain 5/5; analytical 4/5; communication 4/5; planning 3/5; tools 2/5.	Needs automation/API and configuration discipline.
Former developer	Technical 5/5; domain 2/5; analytical 4/5; communication 3/5; planning 3/5; tools 5/5.	Needs independent testing mindset and tariff knowledge.

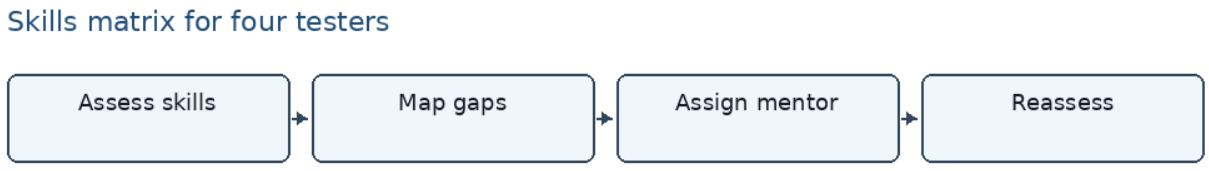


Figure: Skills matrix for four testers (illustrative process diagram).

Recommendation and limitation

Assess each skill using a short practical task, not self-rating alone. The billing specialist should approve business interpretation but should not be the sole reviewer of expected values; the former developer can mentor automation while learning the independence of testing.

TASK 7.2 Strategy A: classroom and certification

A four-week training plan for the two graduates should combine concepts with practical checks. Certification alone does not prove ability to test financial boundaries.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Week 1 – fundamentals	8 hours: test levels, defect lifecycle, equivalence partitioning and boundary values; 2-hour quiz.	
Week 2 – documentation	8 hours: test case, procedure, incident and summary templates; write a report for 500-unit defect.	
Week 3 – tools	8 hours: issue tracker, spreadsheet oracle, API client and basic SQL/log reading.	
Week 4 – assessment	8 hours: execute 12 cases, submit traceability and reproduce one incident independently.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Cost	Course/certification estimate must be quoted; reserve part of ₹1,50,000 tool/training budget only after approval.	
Strength	Consistent terminology and broad foundation.	
Gap	May leave real-world debugging, communication under pressure and domain-specific judgment underdeveloped.	

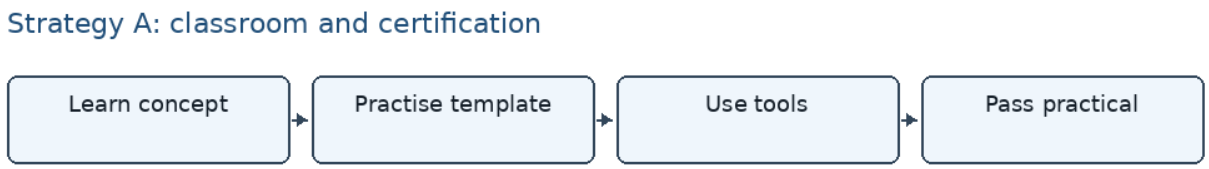


Figure: Strategy A: classroom and certification (illustrative process diagram).

Recommendation and limitation

Use classroom training for baseline vocabulary, not as the only development strategy. Evaluate with practical evidence: correct boundary tests, complete incident, traceability and independently reproduced failure.

TASK 7.3 Strategy B: mentoring and pair testing

Prediction: mentoring will give the graduates the quickest task-specific improvement because they receive immediate feedback while creating real PowerPay artefacts.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Week 1	Pair with experienced tester; observe one bill/payment test session; write observations and questions.	
Week 2	Design cases for 100/101/300/301/500/501 units with mentor review.	
Week 3	Run tests, file incident, use logs and identify root cause with developer shadowing.	

Week 4	Lead a small test set independently; mentor scores correctness, completeness and communication.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Strength	Feedback is contextual; domain knowledge transfers quickly.	
Risk	Mentor style varies; one mentor may pass on bad habits or become a bottleneck.	
Control	Use common rubric, pair rotation and independent review by test manager.	

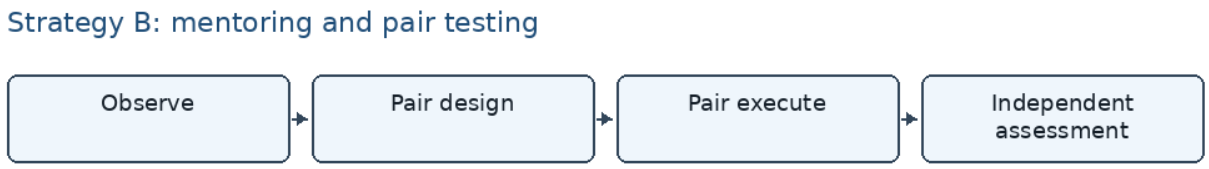


Figure: Strategy B: mentoring and pair testing (illustrative process diagram).

Recommendation and limitation

Pair testing should be the immediate method for critical billing/payment work. It fails if the graduate only watches and never performs, or if the mentor gives answers without asking for evidence. Every week should include an independently completed artefact.

TASK 7.4 Strategy C: skills matrix and rotation

A skills matrix creates visibility of coverage and supports rotation across test design, automation, defect analysis and reporting.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Rotation 1: design	Graduates produce boundary and negative tests; billing specialist reviews domain interpretation.	
Rotation 2: automation	Former developer mentors scripts; reviewer verifies assertions independently.	
Rotation 3: defect analysis	Each trainee reproduces one incident and links evidence to build and requirement.	
Rotation 4: reporting	Each trainee creates dashboard data and a concise risk narrative.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Strength	Balanced capability and backup coverage.	
Risk	Rotation can be too shallow under a one-week assignment or release deadline.	
Measure	Skill score before/after, independently passed tasks and critical-case backup ownership.	

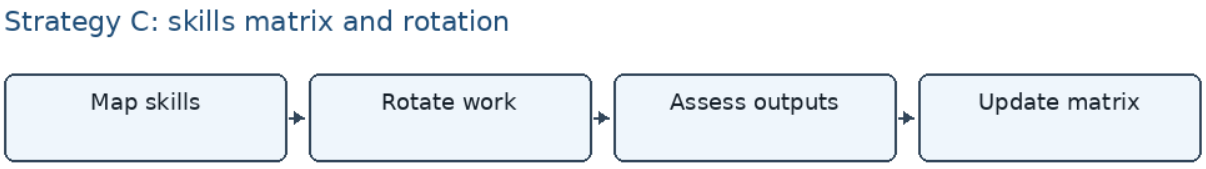


Figure: Strategy C: skills matrix and rotation (illustrative process diagram).

Recommendation and limitation

Use rotation after basic mentoring has established safe practice. Do not rotate a novice into unsupervised release-critical billing approval. The skills matrix should be updated after practical demonstrations, not merely attendance.

TASK 7.5 Four-week plan comparison and cost

Prediction: the best result is a combination—classroom fundamentals, mentoring for high-risk work and rotation for sustainable team resilience.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Classroom	Estimated 32 trainee-hours over four weeks plus instructor time; course fee depends on provider.	Good common base; slower transfer to real work.
Mentoring	About 2 hours/day × 4 weeks per graduate plus reviewer time; mostly internal effort.	Fast practical gains; mentor capacity cost.
Skills matrix/rotation	Setup 4–6 hours plus weekly review and supervised rotation.	Improves coverage; initial coordination effort.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Budget	₹1,50,000 is the tool budget; training spend must be explicitly allocated rather than assumed included.	Track licence and staff hours separately.
Skill gain	Score practical tasks 0–5 before and after.	Report actual scores, not predicted gains.
Operational cost	Lost tester capacity during training must be scheduled.	Protect critical suite ownership.

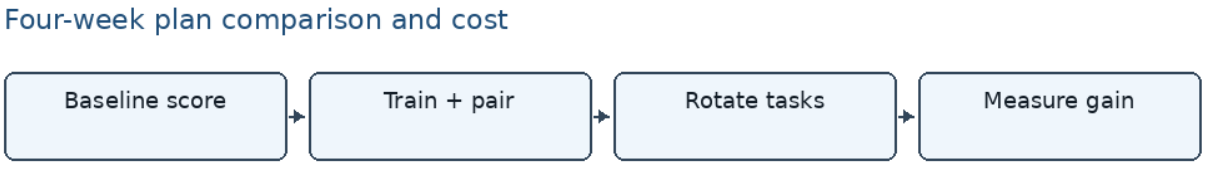


Figure: Four-week plan comparison and cost (illustrative process diagram).

Recommendation and limitation

Combination: week 1 classroom foundations and pairing; weeks 2–3 guided bill/payment execution and automation; week 4 rotation and independent practical assessment. Training succeeds if graduates can execute, explain and document a test—not only pass a quiz.

TASK 7.6 AI changes and essential human skills

AI can accelerate test-data ideas, draft code and summarise logs, but may hallucinate requirements or output expected values derived from the implementation itself.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
AI-supported tasks	Generate candidate boundary values, explain stack traces, draft test skeletons, summarise logs and identify likely code changes.	
Human-only accountability	Approve tariff interpretation, protect personal data, decide severity, assess residual risk and authorise release.	
Validation controls	Require requirement ID, independent expected value, code review, execution evidence and prompt/data policy compliance.	
Bias and blind spots	AI may overfocus on common cases, omit exact boundary values or produce syntactically plausible but incorrect code.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Measure	Compare AI-drafted cases with expert-reviewed case set.	Track omissions and invented cases.
Security	Use synthetic data and approved tools only.	No real consumer records in public prompts.
Skill implication	Critical thinking, domain knowledge, communication and evidence discipline become more important.	Automation does not replace judgment.

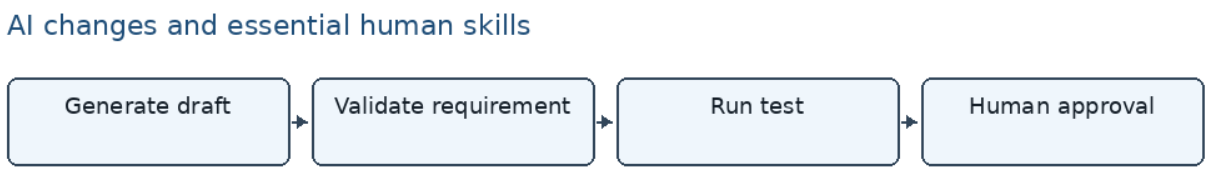


Figure: AI changes and essential human skills (illustrative process diagram).

Recommendation and limitation

AI is an assistant, not a test authority. A tester must be able to explain why a case exists, what result is expected and what evidence proves the outcome. If a generated case lacks a traceable requirement or uses an unverified amount, reject it.

TASK 7.7 Task 7 final recommendation

The team should develop a T-shaped skill profile: all testers share fundamentals, while at least two people can cover each critical area such as tariff boundaries, payment idempotency, automation and reporting.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Training mix	Classroom foundation + mentoring/pair testing + skills matrix and rotation.	
Minimum coverage	At least two named owners and one backup for every critical test suite.	

Assessment	Practical test design, execution, incident reproduction, traceability and dashboard accuracy.	
Reassessment	Monthly and after major tariff/payment changes.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Risk reduced	Boundary faults, poor incidents, slow diagnosis and single points of failure.	
Evidence	Skills matrix before/after, training attendance, practical rubric and assigned case ownership.	
Failure condition	Training is completed but no protected practice time is allocated.	Schedule training within capacity plan.

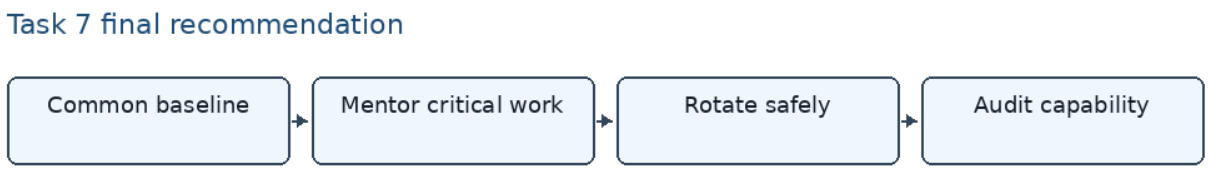


Figure: Task 7 final recommendation (illustrative process diagram).

Recommendation and limitation

Recommendation: mentoring plus practical rotation, supported by short classroom learning. Revisit if the billing specialist is unavailable or if the former developer becomes the only automation owner; cross-train a second person and retain a documented runbook.

TASK 8.1 Integrated process and document set comparison

Prediction: risk-tailored hybrid documentation gives full evidence for billing/payment while avoiding duplicative paperwork for lower-risk modules.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Full IEEE-style set	Plan, design, case spec, procedure, log, incident and summary for every level.	Highest completeness; high authoring and maintenance effort.
B. Lean tool-generated	Tool stores cases, logs, incidents and reports; narrative limited to plan and summary.	Fast and consistent; weak if tool metadata is incomplete or exports are unavailable.
C. Risk-tailored hybrid	Full set for billing/payment; lean generated records for other modules.	Balances auditability and effort according to consumer-money risk.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Completeness	Check every required artefact exists and links to a baseline.	Billing/payment require complete records.
Effort	Estimate authoring + review + maintenance hours by document set.	Compare hours, not vague labels.
Change ease	Change tariff and inspect affected documents/cases.	Traceability should identify impacted artefacts.

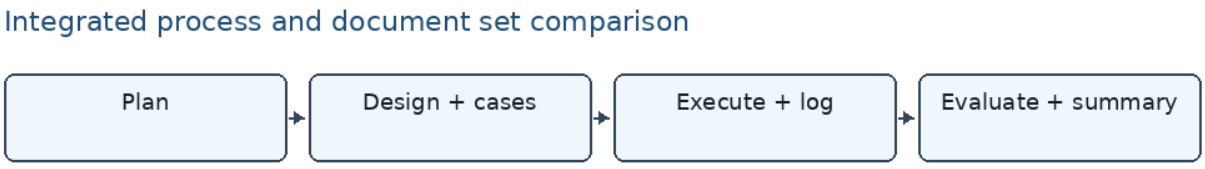


Figure: Integrated process and document set comparison (illustrative process diagram).

Recommendation and limitation

Choose C. The skip rule is: a document may be omitted only if the same information is reliably captured in a controlled tool record, is retrievable for audit, has a named owner, and the risk owner approves the tailoring. No critical billing/payment evidence may be skipped.

TASK 8.2 Full document set for bill generation

The bill feature needs detailed documents because the historical 500-unit error affected money and could recur after tariff changes.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Test plan extract	Scope PP-R04/R05; build/tariff baseline; approach; entry/exit criteria; risks; responsibilities; schedule.	
Test design spec	Boundary analysis, equivalence partitions, independent oracle, negative inputs and rounding.	
Case specification	TC-BILL-001 onward with preconditions, input units, expected breakdown and result.	
Procedure	Verify build, reset synthetic data, upload reading, generate bill, capture breakdown, compare oracle.	
Log / incident	Every attempt with build ID; incident for any mismatch.	
Summary	Coverage, results, deviations, residual risks, release recommendation and approval.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Owner	Test manager coordinates; tester writes; billing coordinator reviews expected amounts.	Clear ownership.
Change control	Tariff revision updates baseline, design, expected data and impacted cases.	Avoid stale expected values.
Effort estimate	Full set estimate: plan 4h, design 3h, 8 cases 4h, procedure 2h, logs 1h, incidents as needed, summary 2h; refine using time log.	Illustrative initial estimate, not actual recorded time.

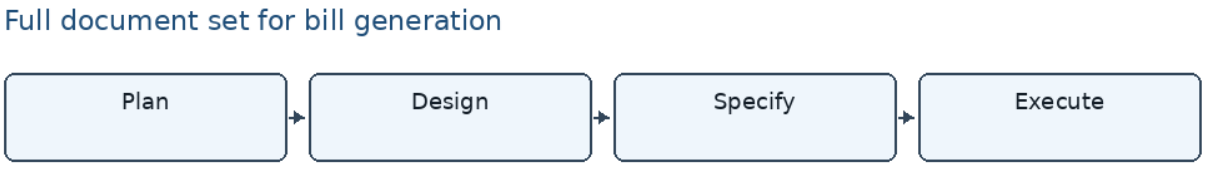


Figure: Full document set for bill generation (illustrative process diagram).

Recommendation and limitation

Use full documents for the bill-generation feature and payment idempotency. Keep cases structured so the tool can generate logs and reports, avoiding duplicate manual copying. Document depth should follow risk, not habit.

TASK 8.3 Lean tool-generated document set

Lean documentation is viable when the tool preserves version, owner, timestamps, result, evidence and change history.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Plan	Short narrative plan and risk register with links to tool objects.	
Cases	Tool record includes requirement ID, precondition, steps, expected result, priority and baseline.	
Execution log	Auto-generated status, timestamp, tester, environment, build and evidence link.	
Incident	Linked issue with reproduction, severity, expected/actual and attachments.	
Summary	Tool export plus reviewed narrative of scope, deviations and residual risk.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Benefit	Less duplication; rapid daily reporting.	Useful for complaint/outage and lower-risk screens.
Risk	Tool migration/export failure or incomplete metadata may break audit trail.	Validate backups and export capability.
Acceptance	Audit sample retrieves every record without private oral explanation.	Tool output is not automatically correct.

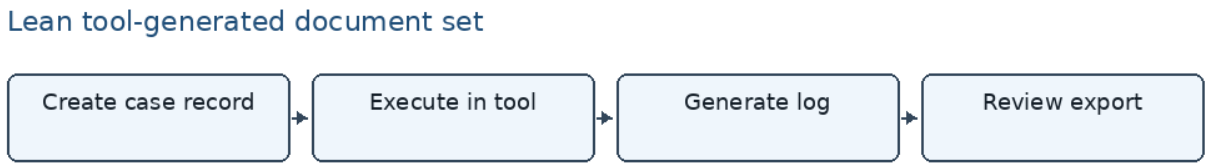


Figure: Lean tool-generated document set (illustrative process diagram).

Recommendation and limitation

Use lean records only if the tool is approved, versioned, backed up and exportable. A dashboard that shows only status without the tested build, requirement and evidence does not satisfy the audit need.

TASK 8.4 PowerPay test-process flow and ownership

The process is sequential for control but iterative in practice: failed tests return to debugging and retest, while approved changes return to impact analysis and design.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication

Plan	Test manager owns plan, schedule, risk register and criteria; Chief Engineer approves.	
Identify items	Build engineer owns manifest; test manager verifies SRS, module versions and testware baseline.	
Design	Tester owns design/cases; billing coordinator approves tariff interpretation; developer reviews interface feasibility.	
Execute	Tester runs procedure and logs; developer diagnoses and fixes; build engineer maintains environment.	
Report	Tool generates logs/dashboard; test manager validates weekly status and level summary.	
Evaluate	Release authority checks critical criteria, residual risks and evidence completeness.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Loop	Failure → incident → debugging → fix build → regression.	Never overwrite original failed run.
Gate	Unidentified build or critical blocker suspends execution.	Resume after smoke and baseline verification.
Output	Approved release decision linked to exact candidate.	No release based on verbal assurance.



Figure: PowerPay test-process flow and ownership (illustrative process diagram).

Recommendation and limitation

Central ownership does not mean central execution. Each role maintains its own artefacts, while the test manager checks completeness and consistency. The flow must show who creates, reviews and approves each document.

TASK 8.5 Effort comparison and auditability

Effort should be estimated per feature and updated with actual authoring/review time. The following estimates are illustrative and show relative burden.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Full IEEE set	Plan 4h + design 3h + cases/procedure 6h + log/incident maintenance 3h + summary 2h = about 18h per feature cycle.	High completeness; repetitive for low-risk functions.
Lean tool-generated	Plan 2h + configure cases 4h + tool execution/report review 2h = about 8h.	Low effort if tool is reliable and metadata is complete.
Risk-tailored hybrid	Billing/payment full set (~18h); lower-risk module lean set (~8h each).	More effort where financial impact is high.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation

Auditability	Check sample of case → result → build → requirement chain.	Hybrid strongest if configuration identity is compulsory.
Readability	Short executive view plus detailed linked evidence.	Avoid long duplicated narrative.
Change effort	Count artefacts affected by tariff change and time to update.	Traceability reduces manual search.

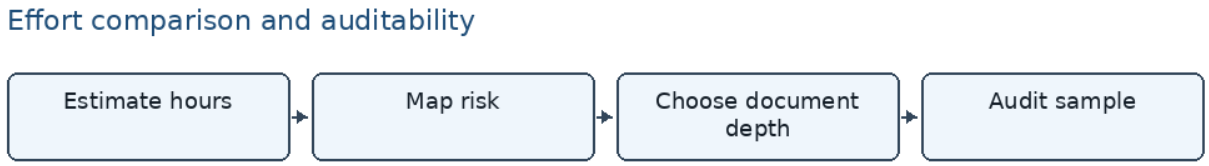


Figure: Effort comparison and auditability (illustrative process diagram).

Recommendation and limitation

Choose hybrid with automated generation for repeatable records. The trade-off is that lean documents may conceal context if the tool fields are poorly designed; audit sampling and exports are mandatory. Revisit if the tool cannot retain complete history.

TASK 8.6 AI-generated documentation and final rule

AI can draft test procedures and summaries from test cases but may invent coverage, omit failed tests or state release readiness unsupported by logs.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Acceptable uses	Draft plain-language procedure, suggest test cases, summarise actual exported results.	
Required checks	Every claim tied to source record; no invented counts; build and environment consistent; no unmasked data.	
Audit restriction	AI output is a draft until reviewed, versioned and approved by an accountable human.	
Skip rule	Skip duplicate narrative only when tool record contains the same required information and is retrievable.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Accuracy	Compare all metrics against raw export.	No unexplained discrepancy.
Completeness	Check requirements, failures, blocked tests, deviations and residual risks.	No critical omission.
Ownership	Reviewer name, date, version and source export recorded.	Creates accountability.

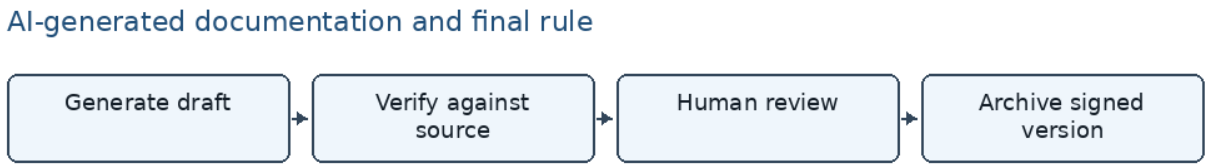


Figure: AI-generated documentation and final rule (illustrative process diagram).

Recommendation and limitation

Final recommendation: risk-tailored hybrid. It addresses the pilot's four failures while controlling paperwork. AI-generated documentation cannot be accepted for audit without review because plausible prose can conceal missing evidence or invent a conclusion.

TASK 9.1 Changed scenario and impact analysis

PowerPay now serves 12 lakh consumers, receives smart-meter uploads every 15 minutes, has four weeks, three testers, two offices, masked-data obligations and mandatory audit evidence. Prediction: retaining the original plan unchanged is unsafe.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Duration	10 weeks → 4 weeks.	Less time for manual design, execution, retest and acceptance.
Team	4 testers → 3 testers.	Reduced capacity and more key-person risk.
Scale	3 lakh → 12 lakh consumers; smart meter every 15 minutes.	Higher data volume, ingestion frequency and load/concurrency risk.
Location	One coordinated team → two offices.	Handoffs, access and timezone/meeting delays; shared repository required.
Data/privacy	Masking and auditable evidence mandatory.	Real personal data must not leak into logs, screenshots or test exports.
Reporting	Full manual paperwork becomes impractical.	Automate evidence capture without reducing critical controls.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Capacity	3 × 4 weeks × 30 hours/week = 360 gross person-hours.	Before meetings, leave, environment wait and other work.
Demand estimate	400 conditions × (0.4 design + 0.25×3 execution) = 460 hours before retest/overhead.	Already exceeds gross capacity.
Shortfall	Retest 20% of 460=92; subtotal 552; overhead 15%=82.8; total=634.8 hours; shortfall=274.8 hours against 360 gross.	Effective gap is larger after non-test time.

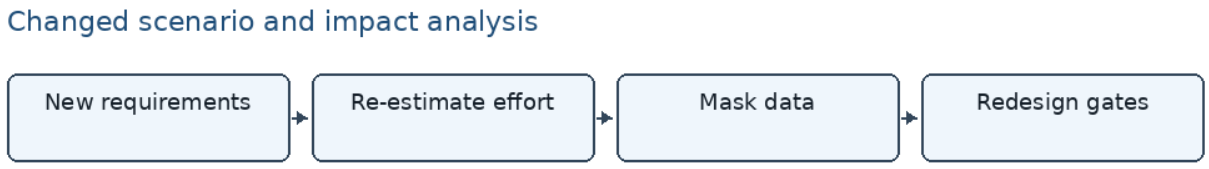


Figure: Changed scenario and impact analysis (illustrative process diagram).

Recommendation and limitation

Redesign is required. The arithmetic gives an illustrative total of 634.8 hours for 400 conditions under the original formula, exceeding gross capacity by 274.8 hours. Do not solve this by silently dropping high-risk billing/payment tests; automate repeatable work, use risk selection, add capacity or adjust release scope/date.

TASK 9.2 Data masking and audit evidence

The privacy constraint applies to test datasets, logs, screenshots, defect reports, dashboard exports, prompts and attachments—not only database rows.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Direct identifiers	Name, phone, address, consumer number and email.	Replace with synthetic IDs or deterministic tokens.
Meter and location data	Meter serial, precise address/location and usage pattern if linkable.	Use generated values and coarse test zones.
Payment data	Bank/account/card identifiers, gateway tokens and transaction secrets.	Use sandbox tokens; never store secrets in incident screenshots.
Evidence	Logs, screenshots, CSV exports, AI prompts and CI artifacts.	Scan and redact before upload; restrict access and retention.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Masking method	Synthetic data preferred; deterministic tokenisation only where relationship testing is necessary.	Keep mapping vault separate and access-controlled.
Audit record	Test ID, build, environment, timestamp, result, masked dataset version and evidence hash.	Proves result without exposing identity.
Validation	Automated secret/PII scan plus human spot check before report publication.	Block export if sensitive data detected.



Figure: Data masking and audit evidence (illustrative process diagram).

Recommendation and limitation

Privacy control must be built into the process before testing begins. A report that masks the main table but leaves phone numbers in a screenshot or log is still unsafe. If masking changes data relationships, verify the synthetic dataset preserves the conditions under test.

TASK 9.3 Original process: valid versus unsafe

The original process remains a foundation but its capacity assumptions, reporting frequency and data practices must be reassessed.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Still valid	Risk-based testing, independent expected values, baseline identity, regression, critical exit gate and human acceptance.	Core principles survive scale change.
Needs modification	10-week schedule, four-tester allocation, 180-condition estimate, one-office communication and manual reports.	These assumptions no longer match the project.

Unsafe unchanged	Unmasked production-like data, stale 3-lakh load profile, same pass-rate threshold despite expanded risk, and release build not verified.	May create privacy incident or miss peak load/financial defects.
Exit criteria	Keep zero critical billing/payment/privacy defects and exact build identity; strengthen coverage/evidence rules.	Do not lower critical integrity threshold to meet four weeks.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Checklist	Mark valid / modify / unsafe and assign action owner.	Review at daily redesign meeting.
Least trusted metric	Raw overall pass percentage without risk weighting and denominator.	Can hide untested smart-meter paths and critical defects.
Reapproval	Chief Engineer approves changed scope, capacity, criteria and residual risks.	Original approval does not cover new conditions.

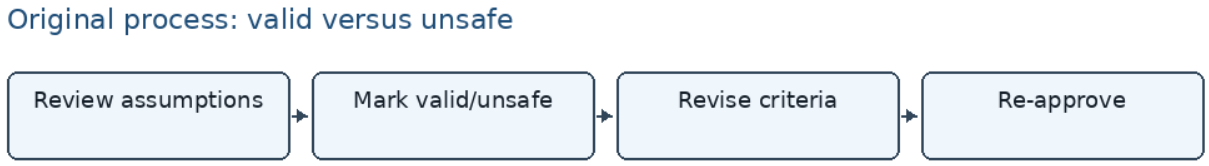


Figure: Original process: valid versus unsafe (illustrative process diagram).

Recommendation and limitation

Retain sound principles but replace the plan baseline. A 95% pass rate across a narrow suite is not evidence of readiness for 12 lakh consumers. Report critical-risk coverage, load profile, data masking status and evidence completeness alongside counts.

TASK 9.4 Compare A/B/C adaptation strategies

Prediction: strategy C will best satisfy audit and privacy obligations, with B's iteration cadence and embedded testers as implementation mechanisms.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
A. Retain and cut scope	Quick to start; reduces breadth but preserves old staffing and document assumptions.	High risk of inadequate load, privacy and evidence coverage.
B. Modify process	Short iterations, embedded testers in both offices, shared tool and live reporting.	Improves coordination and speed; may leave policy and central audit controls inconsistent.
C. Redesign organisation/policy	Central test group, lightweight standard records, automated evidence capture and revised release gates.	Strongest audit consistency; setup effort and governance overhead.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Feasibility	Compare planned hours with 360 gross hours and available automation.	A fails without substantial risk acceptance.
Audit	Check each result has build, requirement, masked data version and timestamp.	C strongest if tool controls are implemented.
Team effectiveness	Measure handoff delay, duplicate work and critical case ownership.	B helps distributed teams; C defines standards.

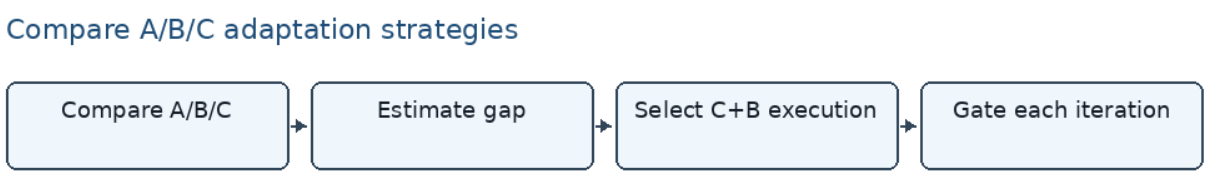


Figure: Compare A/B/C adaptation strategies (illustrative process diagram).

Recommendation and limitation

Choose C as the control model, implemented through B's short iterations and embedded module testers. This is a redesign, not simply a documentation reduction. If tooling cannot be ready, use controlled CSV/Word templates with signed manifests as a temporary fallback.

TASK 9.5 Revised four-week plan outline

A four-week plan must front-load requirements clarification, environment readiness, masking and automation. Testing should not wait until week four to discover a wrong tariff baseline.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Week 1	Freeze SRS/tariff version; mask/synthesise data; establish CI baseline, contract tests, risk ranking and test plan; smoke test environment.	
Week 2	Execute critical billing boundaries, payment idempotency, meter ingestion validation, access/privacy and core end-to-end tests.	
Week 3	Load/rate tests for 15-minute ingestion and consumer scale; integration/recovery; regression and defect fixes.	
Week 4	Run critical regression on release candidate; verify audit evidence, retest fixes, acceptance review and go/no-go.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Daily gate	Build identity verified; critical cases executed; blockers and privacy scan status reported.	No stale build.
Parallelism	Developers own unit tests; testers focus on independent integration/system/risk cases.	Protect scarce tester capacity.
Contingency	Late tariff change or failed critical gate triggers release hold or scope/date negotiation.	No silent criterion relaxation.

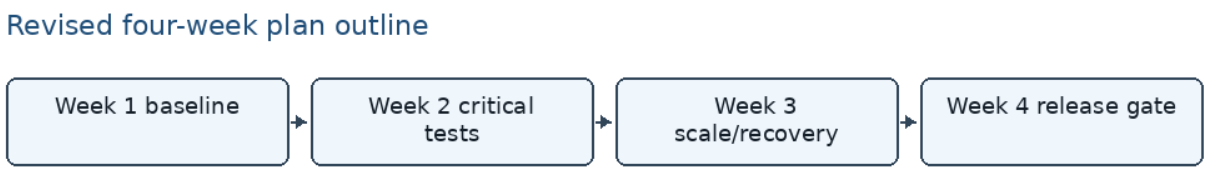


Figure: Revised four-week plan outline (illustrative process diagram).

Recommendation and limitation

Sequence the plan by dependencies: approved tariff and masked data precede valid execution; load environment must be available before performance evidence; release candidate must be frozen before final regression and summary. Any major tariff change resets impacted evidence and may invalidate the four-week date.

TASK 9.6 Revised exit criteria and daily dashboard

Exit criteria must control risk rather than reward activity volume. A short schedule is not a reason to accept unknown critical outcomes.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Critical integrity	Zero open critical defects for bill calculation, duplicate payment, payment/receipt consistency, privacy or tested-build identity.	
Coverage	100% of critical requirements mapped and executed on candidate build; all unexecuted high-risk cases explicitly listed with approved disposition.	
Regression	All tariff boundary and payment idempotency regressions pass on exact release candidate.	
Load	Approved workload reaches defined target; report p90, throughput, error rate and resource bottlenecks. Target values must be approved before test.	
Privacy	No unmasked personal data in test data, logs, screenshots, reports or AI prompts; scan evidence retained.	
Audit	Every result includes requirement/test ID, build, environment, masked dataset version, timestamp, status and evidence link.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Daily dashboard	Cases planned/executed/pass/fail/blocked; critical risks; defects by severity/age; changed requirements; build ID; masking scan; load progress.	Send to Chief Engineer daily.
Readiness statement	Ready only if all mandatory gates pass or authorised residual-risk decision is recorded where policy permits.	No green status based on average pass rate.
Trend	New/closed/reopened critical defects and test execution by risk.	Explains whether quality is improving.

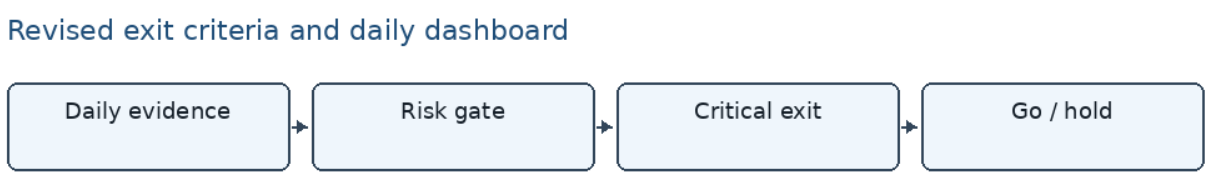


Figure: Revised exit criteria and daily dashboard (illustrative process diagram).

Recommendation and limitation

Use a two-layer gate: critical mandatory conditions cannot be traded for speed; lower-risk breadth can be risk-ranked and deferred with documented approval. If load tests are not representative of smart-meter bursts, state that the system has not been demonstrated ready for that load.

TASK 9.7 Evidence-based defence of redesign

The final choice follows from the capacity calculation, earlier pilot failures, risk ranking and privacy obligation—not from a preference for new tools.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Effort	Estimated need 634.8 h vs 360 gross h; minimum shortfall 274.8 h before overhead from meetings and waiting.	Automation and scope negotiation required.
Historical defects	500-unit error and 18% reopen rate indicate boundary and fix-verification weakness.	Keep independent oracle and regression.
Payment integrity	Duplicate record indicates idempotency/reconciliation risk.	Keep retry/timeout/duplicate callback tests mandatory.
Configuration	Wrong tested build indicates baseline failure.	Hash and release-manifest gate remains mandatory.
Privacy/audit	New law requires masked data and auditable results.	Add privacy scan and machine-linked evidence.

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Trade-off	Less full narrative for low-risk cases; more automated standard records.	Potential loss of context if tool exports are poor.
Failure condition	Late tariff revision after regression freeze or unavailable load environment.	Reassess scope/date, rerun impacted critical cases, hold release if evidence incomplete.
Next change	Increase capacity, reduce noncritical scope, or move release date; never waive core financial/privacy gate.	Decision owner: Chief Engineer.

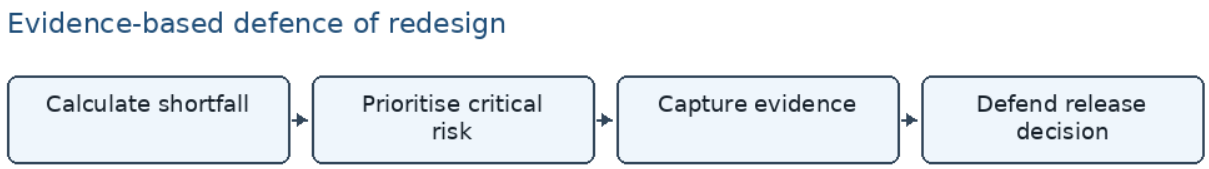


Figure: Evidence-based defence of redesign (illustrative process diagram).

Recommendation and limitation

Defend redesign with explicit evidence. It gives up some documentation verbosity and breadth in low-risk areas in exchange for speed, but preserves full billing/payment traceability and privacy controls. If a late tariff change invalidates expected values, re-baseline and rerun; if there is insufficient time, delay release rather than ship unverified financial logic.

TASK 9.8 Final revised flowchart and ownership

This page consolidates the redesigned process and the accountable owner for each stage.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Plan and mask	Test manager + privacy/data owner: scope, risks, masked data and capacity.	
Baseline and design	Build engineer + testers + billing coordinator: tagged build, tariff hash, test design and oracle.	
Execute in iterations	Embedded testers + developers: automated unit/contract/regression and targeted manual tests.	

Report continuously	Tool: logs and evidence; test manager validates daily dashboard; defects triaged across both offices.	
Evaluate and release	Chief Engineer/release board: exit criteria, residual risks, audit evidence and go/hold decision.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Ownership	Every artefact has creator, reviewer and approver.	No anonymous sign-off.
Distributed coordination	Shared repository, common test IDs, asynchronous updates and daily cross-office triage.	Avoid conflicting copies.
Fallback	If shared tool fails, use versioned controlled templates and restore evidence after recovery.	Do not stop audit trail.

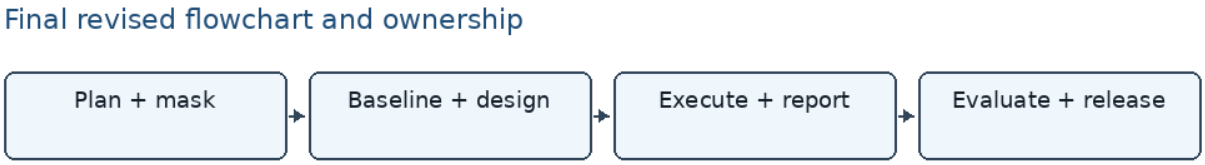


Figure: Final revised flowchart and ownership (illustrative process diagram).

Recommendation and limitation

Recommended final model: redesigned central policy and evidence model, iterative execution, module-embedded testers, risk-tailored documentation, automated evidence capture and a formal internal release gate. The process should be reviewed after the first release using actual defect escape, reopen rate, audit completeness and delivery time.

TASK 9.9 Task 9 conclusion and measurable follow-up

A redesign should be evaluated after release; otherwise the Corporation cannot tell whether it improved outcomes or merely changed tools.

Comparison of the three techniques

Technique	Strengths / limits	PowerPay implication
Metric 1	Critical defect escape rate for billing/payment; review each release and post-release period.	
Metric 2	Reopen rate compared with pilot baseline of 18%, using a consistent definition and window.	
Metric 3	Percentage of test results with complete build/requirement/data/timestamp evidence; target 100%.	
Metric 4	Privacy scan findings in test artefacts; target zero exposed personal records.	
Metric 5	Critical risk coverage and execution on the release candidate; target 100% mapped and executed or explicitly held.	

Applied evidence / decision measures

Measure	Method / evidence	Interpretation
Review cadence	After release and monthly for first quarter.	Assign owners and due dates.

Improvement	Investigate missed controls, not only individual errors.	Feed lessons into policy and training.
Revisit trigger	Major architecture change, tariff regulation change, data-law change or repeated critical escape.	Reopen plan and capacity estimate.

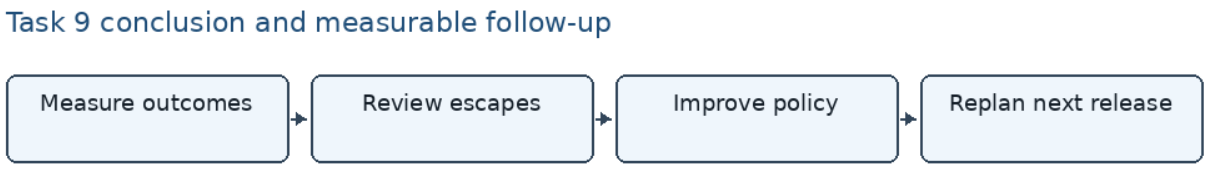


Figure: Task 9 conclusion and measurable follow-up (illustrative process diagram).

Recommendation and limitation

Task 9 conclusion: retain principles that protect financial correctness, revise the assumptions that no longer fit, and redesign the execution and evidence system for scale, short schedule and privacy. A safe release is an evidence-based decision, not a deadline-driven declaration.

APPENDIX APPENDIX A – consolidated risk register

The register consolidates risks across the original and changed scenarios. Probability and impact are planning estimates on a 1–5 scale; the project team must review them with evidence at kickoff and each weekly gate.

Risk	P	I	Score	Mitigation / contingency	Owner
Wrong tariff boundary	3	5	15	Approved oracle; boundary suite; tariff versioning; block release on critical mismatch.	Billing coordinator + tester
Duplicate payment	3	5	15	Idempotency key; duplicate callback and timeout tests; reconciliation.	Payment developer + tester
Wrong build/file	4	5	20	Immutable manifest, hash verification and release candidate freeze.	Build engineer
Test server unavailable	4	5	20	Booking calendar, smoke checks, backup image, escalate after half-day.	Build engineer
Late tariff change	3	5	15	Change request, impact analysis, reapproval and critical regression.	Test manager
Tester absence	3	3	9	Cross-training and backup ownership.	Test manager
Privacy leakage	2	5	10	Synthetic/masked data, scans and access controls.	Data owner
Reopened defects	3	4	12	Fix review, regression evidence and root-cause analysis.	Development lead
Smart-meter burst overload	4	5	20	Load profile, queue monitoring, soak/recovery tests.	Performance owner

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX B – representative bill and payment cases

Use the following as a starter suite. Numeric expected bills must be populated from the approved tariff semantics and tax/rounding policy, not inferred from the implementation.

ID	Requirement	Input / event	Expected result	Priority
PP-TC-001	R04	99 units	Correct approved slab total and breakdown.	Critical
PP-TC-002	R04	100 units	First boundary matches approved oracle.	Critical
PP-TC-003	R04	101 units	Next slab starts exactly as defined.	Critical
PP-TC-004	R04	299/300/301 units	Adjacent boundary set passes.	Critical

PP-TC-005	R04	499/500/501 units	No off-by-one at historical failure point.	Critical
PP-TC-006	R05	Bill components sum to total	Displayed breakdown equals stored bill total.	High
PP-TC-007	R07	Repeat same payment key	Only one charge/ledger record.	Critical
PP-TC-008	R08	Gateway timeout then callback	One reconciled final status; no duplicate charge.	Critical
PP-TC-009	R09	Confirmed success	Unique receipt with correct amount/reference.	High
PP-TC-010	R10	SMS provider unavailable	Queued/retried or controlled failure; no false receipt claim.	Medium
PP-TC-011	R03	Duplicate meter upload	Duplicate rejected or idempotently handled per specification.	High
PP-TC-012	R14	Unauthorised staff action	Denied and audited.	High

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX C – test log and incident template

Use one row per execution attempt. Re-runs create new records and retain the previous outcome; they do not overwrite failed history.

Field	Required content	Why it matters
Run ID	Unique execution batch identifier	Groups related tests.
Test ID / requirement	Stable case and SRS IDs	Traceability.
Build / config	Build tag, commit, tariff hash, environment ID	Proves what was tested.
Input / data version	Synthetic dataset ID and boundary input	Reproduction without exposing consumers.
Expected / actual	Approved expected result and observed output	Shows the failure precisely.
Status	PASS / FAIL / BLOCKED / NOT RUN	Avoids ambiguous progress.
Evidence	Log/screenshot/report link and hash	Auditable artefact.
Incident ID	Linked issue for failure or blocker	Supports defect lifecycle.
Owner / timestamps	Tester, start/end, reviewer and review date	Accountability.
Fix and retest	Commit ID, regression ID, before/after status	Prevents unverified closure.

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX D – test plan approval checklist

The test manager should complete this checklist before the Chief Engineer's approval meeting. Any missing critical item must have an owner and due date; missing build identity or critical exit criteria blocks approval.

Component	Check question	Status / evidence
Objective and scope	Are features in/out and test objectives explicit?	
Test items	Are build, modules, tariff file and environment identified?	
Approach	Are test levels, techniques, priorities and tools justified?	
Pass/fail criteria	Are item-level and overall exit criteria measurable?	
Suspension/resumption	Are triggers and restart conditions explicit?	
Deliverables	Are plan, cases, logs, incidents and summary assigned?	
Tasks and responsibilities	Are owners, reviewers and approvers named?	

Staffing/training	Does capacity include leave, meetings and mentoring?	
Schedule	Are dependencies and contingency time shown?	
Risks	At least eight risks, owners and contingencies?	
Approvals/change control	Are version history and reapproval rules defined?	

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX E – traceability matrix excerpt

This matrix demonstrates mapping from the authored SRS excerpt to test designs. Expand it to every detailed case and maintain the mapping in the approved test-management tool.

Requirement	Test IDs / design	Level	Critical?	Coverage status
PP-R01 registration	REG-01..REG-05	System/security	No	Mapped
PP-R02 login	AUTH-01..AUTH-08	Unit/system/security	High	Mapped
PP-R03 meter upload	MTR-01..MTR-12	Integration/system/load	High	Mapped
PP-R04 tariff	BILL-01..BILL-20	Unit/system/regression	Yes	Mapped; boundary cases mandatory
PP-R05 bill breakdown	BILL-21..BILL-28	System/reconciliation	High	Mapped
PP-R06 bill uniqueness	BILL-29..BILL-32	Integration	High	Mapped
PP-R07 payment idempotency	PAY-01..PAY-12	Integration/system	Yes	Mapped
PP-R08 payment reconciliation	PAY-13..PAY-20	Integration/recovery	Yes	Mapped
PP-R09 receipt	REC-01..REC-05	System	High	Mapped
PP-R10 SMS	SMS-01..SMS-06	Integration	Medium	Mapped
PP-R11 complaint	CMP-01..CMP-08	System	Medium	Mapped
PP-R12 outage	OUT-01..OUT-06	System/integration	Medium	Mapped
PP-R13 load	LOAD-01..LOAD-10	Performance	High	Mapped; target approval required
PP-R14 access/audit	SEC-01..SEC-10	Security/audit	High	Mapped

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX F – skills matrix and four-week assessment

Scores below are proposed baseline scores based on the limited profiles supplied in the assignment. Confirm with practical assessment before using them as personnel performance ratings.

Skill (0–5)	Graduate 1	Graduate 2	Billing specialist	Former developer	Assessment evidence
Test design	2	2	4	4	Boundary and negative case exercise
Tariff domain	1	1	5	2	Explain slab and 500-unit expected result
Automation/API	2	2	2	5	Create/run regression test
Defect analysis	2	3	4	4	Reproduce incident from report

Communication	3	2	4	3	Triage summary and incident report
Planning/reporting	2	2	3	3	Traceability and dashboard task
Configuration control	2	2	2	4	Verify build/tariff hash
Privacy practice	2	2	3	3	Find and mask sample identifiers

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX G – tool budget and effort control

The ₹1,50,000 budget is a ceiling, not a reason to buy every tool. Prefer existing approved systems and reserve money for configuration, training, backup and support.

Item	Approach	Planning allocation (₹)	Control
Test management	Evaluate existing tool / open-source option first	35,000	Approve only after export and audit trial
CI / automation support	Use existing version control and CI where available	30,000	No duplicate tool if existing capability works
Performance tooling	Open-source load tool plus environment support	25,000	Budget effort and compute separately
Training	Practical training and mentoring materials	20,000	Measure skill gain
Backup / support / contingency	Licence, setup, storage or unexpected need	40,000	Release only with test manager approval
Total ceiling	Illustrative allocation	150,000	Actual quotes and internal costs must be confirmed

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.

APPENDIX APPENDIX H – glossary, assumptions and sign-off

This final appendix defines recurring terms and leaves a controlled place for approvals. Sign-off means the reviewers accept the plan and evidence; it does not mean every risk has disappeared.

Term	Definition for this report
Test item	A build, module, configuration, document or other item selected for testing.
Baseline	An approved, versioned set of artefacts used as the reference for a test cycle.
Traceability	Links among requirements, risks, test designs, cases, results, defects and builds.
Incident	A documented event where observed behaviour differs from expectation or execution is blocked.
Regression test	A repeatable test used to detect whether a change reintroduced a known fault.
Idempotency	Repeating the same payment operation with the same key does not create an additional charge.
Exit criterion	A measurable condition required before a test level or release can be considered complete.
Residual risk	Risk remaining after planned controls and testing, explicitly documented for decision.
Synthetic data	Artificial test data not derived from identifiable consumer records.
Test manager	Name/signature: _____ Date: _____
Billing coordinator	Name/signature: _____ Date: _____
Build engineer	Name/signature: _____ Date: _____
Chief Engineer / release authority	Decision: GO / HOLD Signature: _____ Date: _____

Control note: these artefacts are templates/examples for the assignment. Complete them with approved PowerPay requirements and actual execution evidence before presenting them as project records.